

平坦バンドとディラック点を有する α - T_3 格子におけるカイラルd波超伝導

阪大D2 梶 昌孝

概要

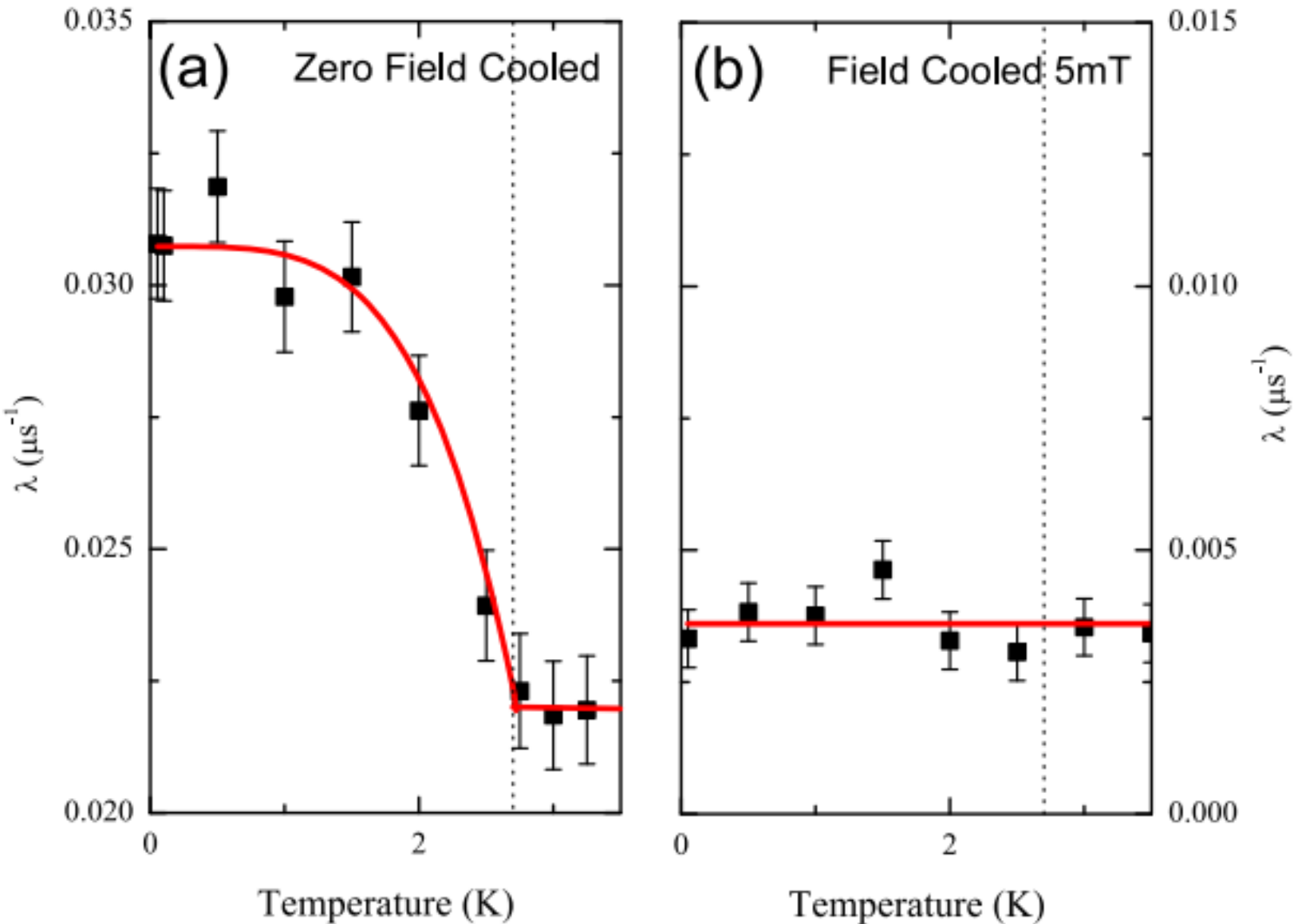
We study superconductivity driven by electron correlations in the nearly quarter-filled α - $\mathcal{T}_3$ lattice. We analyze an extended Hubbard model with off-site attractive interactions within the mean-field framework and find two distinct chiral $d+id'$ -wave superconducting phases characterized by different Chern numbers. We further investigate the superconducting mechanism mediated by spin fluctuations arising from purely repulsive interactions by applying the fluctuation-exchange approximation to the Hubbard model. The gap symmetry obtained by solving the linearized Eliashberg equation is d -wave, corresponding to a $d+id'$ -wave superconducting state characterized by a Chern number of 8. The $\mathbf{q} = \mathbf{0}$ antiferromagnetic spin fluctuation, which possesses the largest spectral weight at finite energies arising from the incipient flat band, gives rise to an effective interaction that favors spin-singlet pairing between sublattice sites that are not directly connected.

自発的時間反転対称性の破れをともなう超伝導

S. K. Ghosh *et al.*, J. Phys.: Condens. Matter 33 (2021)

Table 2. A list of those strongly correlated SCs where μ SR or the Kerr effect have been used to provide evidence for either the presence or absence of TRS in the superconducting state.

Material	Broken TRS?	Reference
UPt ₃	✓/×	[22, 38, 44]
UBe ₁₃	×	[32]
U _{1-x} Th _x Be ₁₃ (0.02 ≤ x ≤ 0.04)	✓	[32]
URu ₂ Si ₂	✓	[23]
UTe ₂	✓	[41]
CeCu ₂ Si ₂	×	[45]
CeCoIn ₅	×	[46]
CeIrIn ₅	×	[46]
YBa ₂ Cu ₃ O ₇	×	[47, 48]
Bi ₂ Sr _{2-x} La _x CuO _{6+δ}	×	[47, 49]
Ba _{1-x} K _x Fe ₂ As ₂	×, ✓ (0.7 ≲ x ≲ 0.85)	[50–52]
Sr ₂ RuO ₄	✓	[21, 33]
Pr(Os _{1-x} Ru _x) ₄ Sb ₁₂	✓	[53, 54]
Pr _{1-y} La _y Os ₄ Sb ₁₂ (y < 1)	✓	[54]
Pr _{1-y} La _y Pt ₄ Ge ₁₂ (y < 1)	✓	[55, 56]



Lorentzian relaxation rate λ_{ZF} obtained from fitting μ SR spectra of LaNiC_2 in (a) zero-field, and (b) a longitudinal field of 5 mT.

カイラル超伝導の最近の話題：菱面体グラフェン

Article

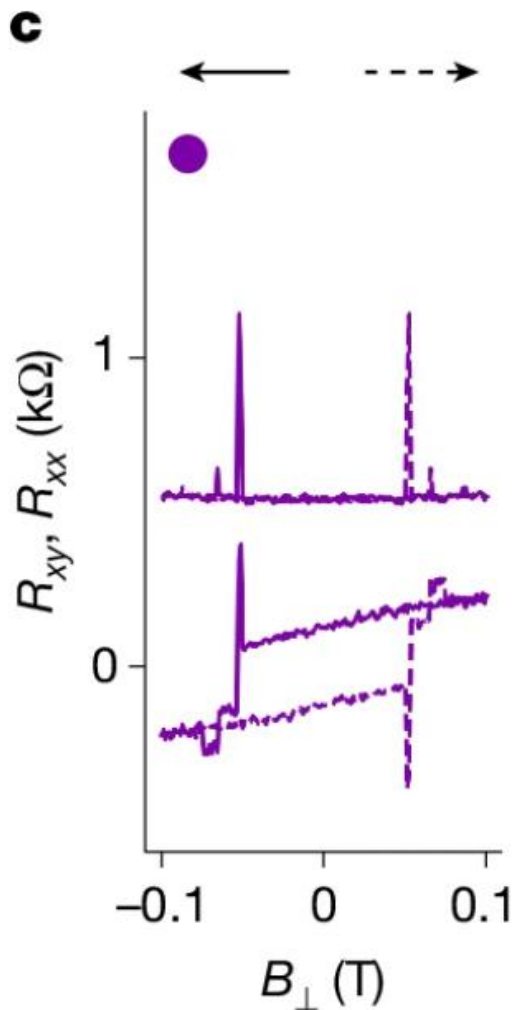
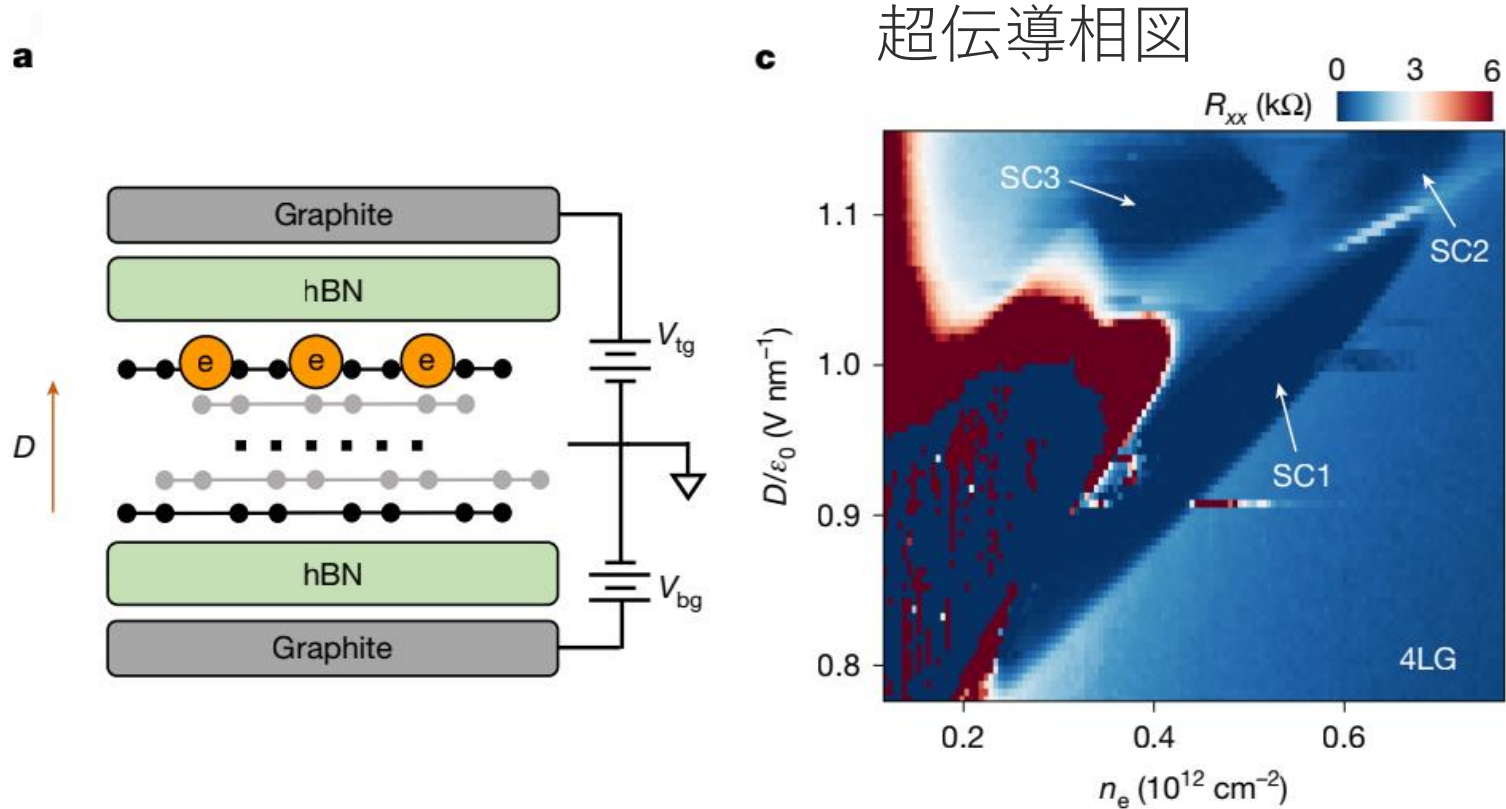
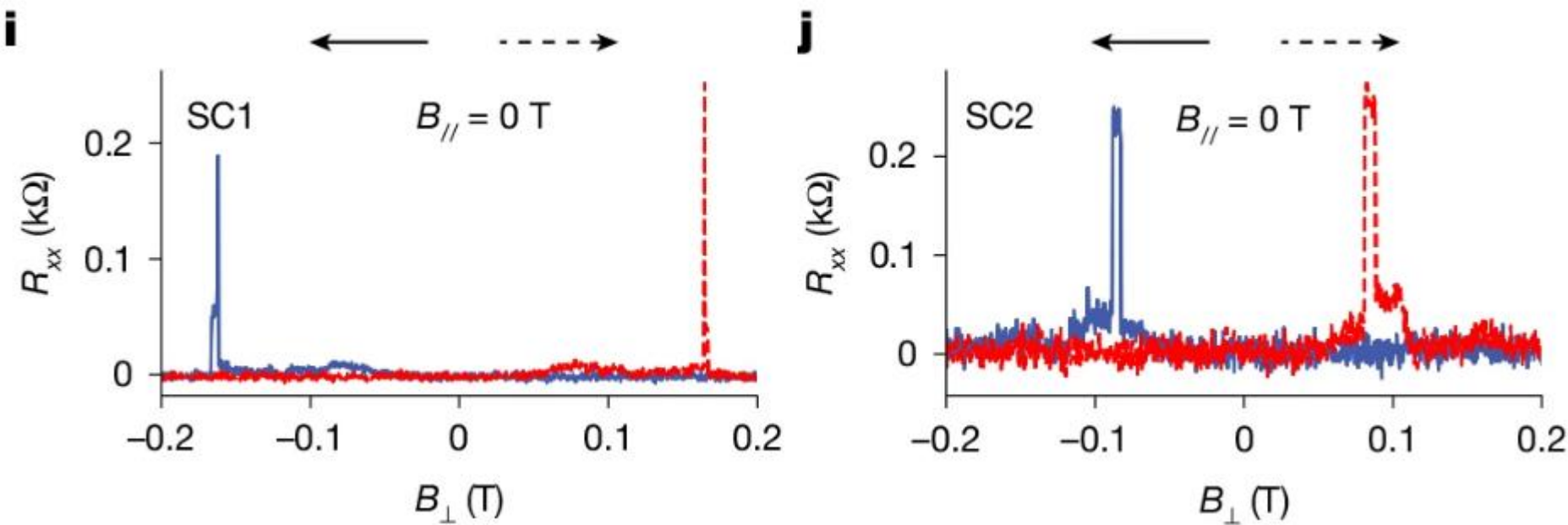
Signatures of chiral superconductivity in rhombohedral graphene

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T. Han *et al.*, Nature 643, 654 (2025).

超伝導下における磁場走査による抵抗のヒステリシス



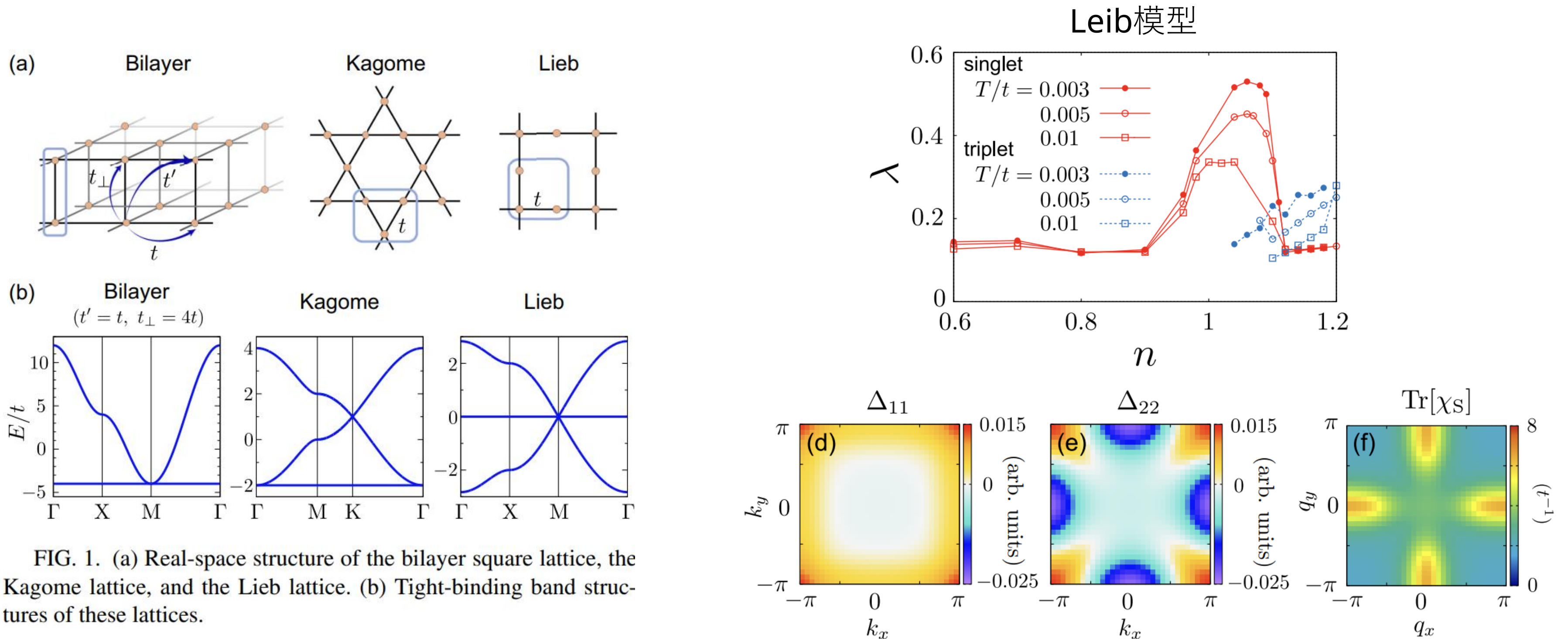
超伝導に隣接する
常伝導相における
異常ホール効果と
ヒステリシス

平坦バンド近傍の超伝導

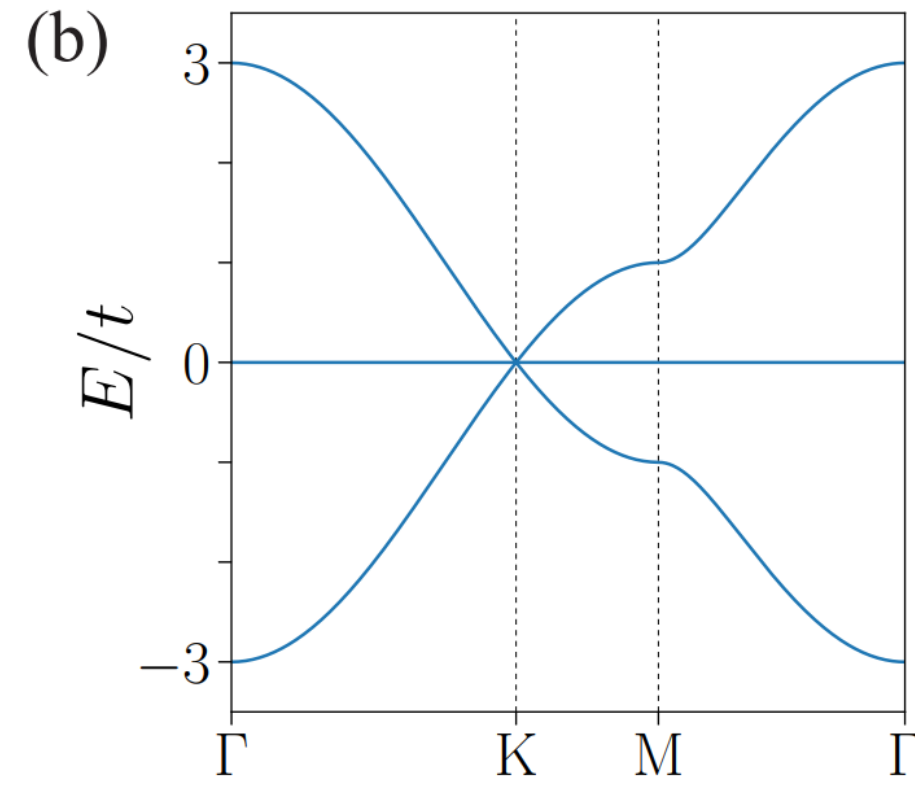
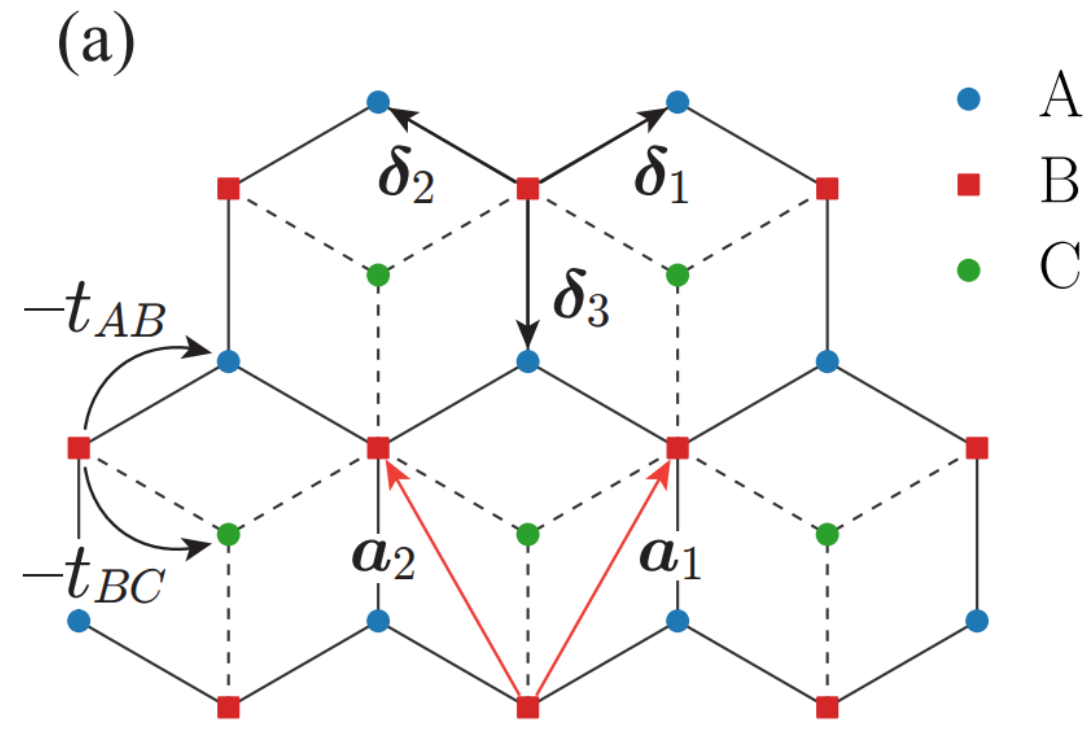
Theoretical study of spin-fluctuation-mediated superconductivity in two-dimensional Hubbard models with an incipient flat band

Tetsuaki Aida,¹ Karin Matsumoto,^{1,*} Daisuke Ogura,^{1,†} Masayuki Ochi^{1,2} and Kazuhiko Kuroki¹

T. Aida *et al.*,
Phys. Rev. B **110**,
054516 (2024)



α - T_3 模型



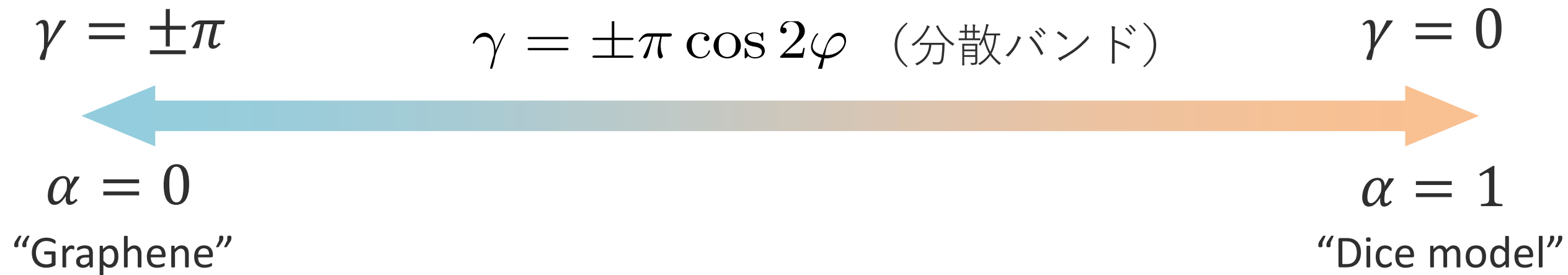
A. Raoux *et al.*, Phys. Rev. Lett. **112**, 026402 (2014).
 Acknowledgements: Prof. F. Piechon

$$t_{ij} = \begin{cases} t_{AB} = t \cos \varphi, & \text{A-B nearest neighbor,} \\ t_{BC} = t \sin \varphi, & \text{B-C nearest neighbor,} \\ 0, & \text{otherwise.} \end{cases}$$

$$H_0(\mathbf{k}) = \begin{pmatrix} 0 & f(\mathbf{k}) \cos \varphi & 0 \\ f^*(\mathbf{k}) \cos \varphi & 0 & f(\mathbf{k}) \sin \varphi \\ 0 & f^*(\mathbf{k}) \sin \varphi & 0 \end{pmatrix} \quad f(\mathbf{k}) = -t \sum_{\ell=1}^3 e^{i\mathbf{k} \cdot \boldsymbol{\delta}_\ell}$$

α - T_3 模型

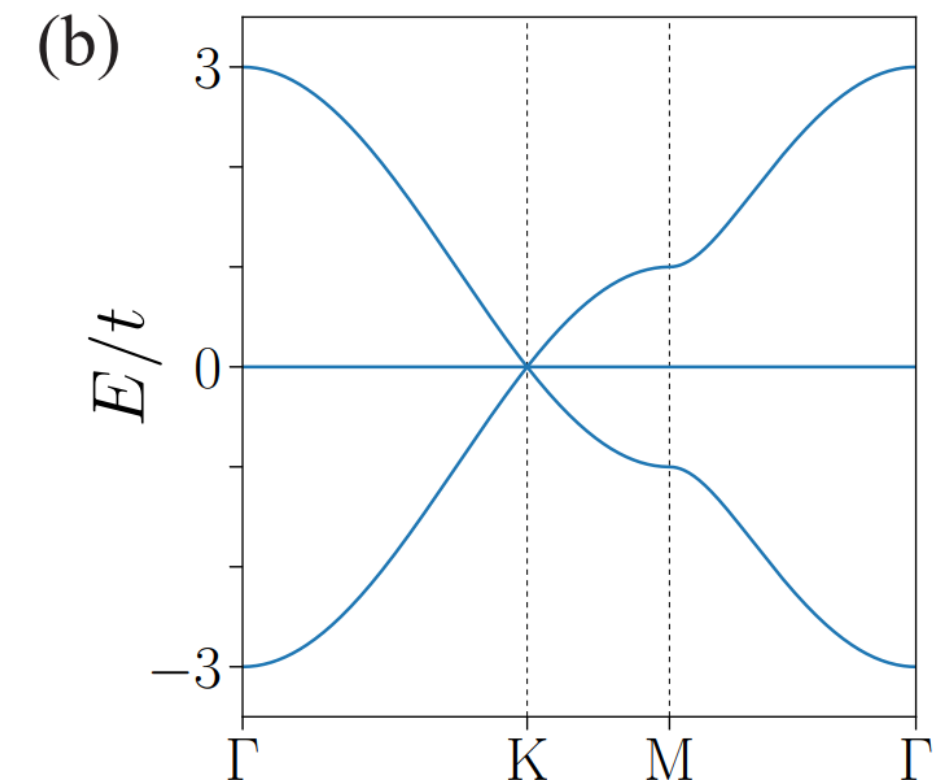
K(K')点を囲む経路に対するBerry位相



ホッピングの比が変わると
固有状態は変化・バンドは不変

固有ベクトル

$$|\mathbf{k}, \pm\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\theta(\mathbf{k})} \cos \varphi \\ \pm 1 \\ e^{-i\theta(\mathbf{k})} \sin \varphi \end{pmatrix}, \quad |\mathbf{k}, 0\rangle = \begin{pmatrix} e^{-i\theta(\mathbf{k})} \sin \varphi \\ 0 \\ -e^{i\theta(\mathbf{k})} \cos \varphi \end{pmatrix}$$



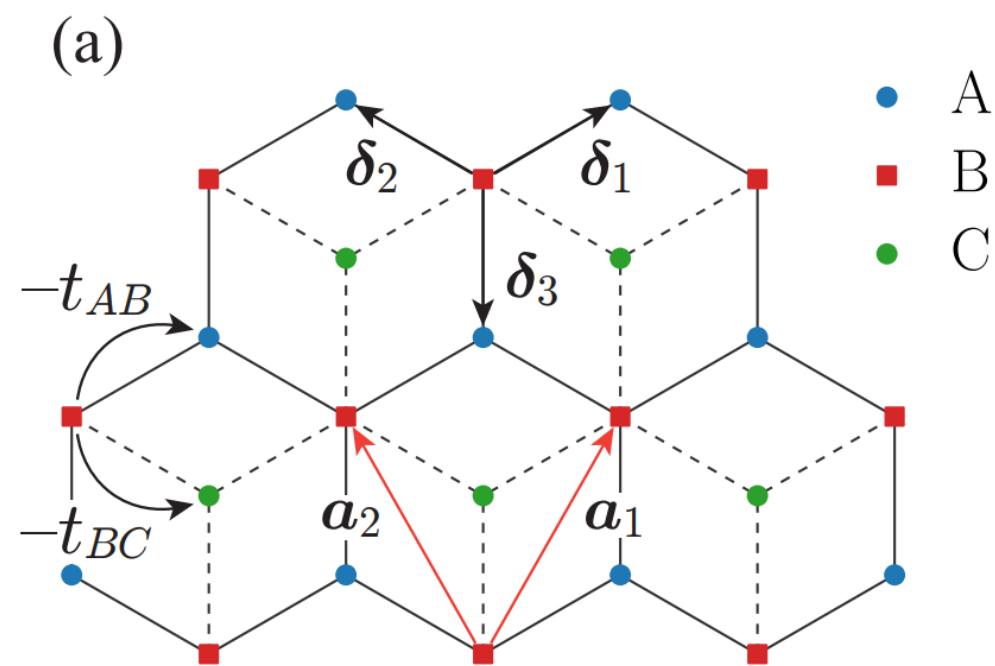
隣接サイト間引力を仮定した拡張ハバード模型の平均場解析

隣接サイトに引力相互作用を加えた拡張ハバード模型

$$\hat{H}_{\text{int}} = U \sum_i \hat{n}_{i,\uparrow} \hat{n}_{i,\downarrow} + \frac{1}{2} \sum_{\langle i,j \rangle, \sigma, \sigma'} V_{ij} \hat{n}_{i,\sigma} \hat{n}_{j,\sigma'} \quad V_{ij} = \begin{cases} -V_{AB}, & \text{A-B nearest neighbors,} \\ -V_{BC}, & \text{B-C nearest neighbors,} \\ -V_{CA}, & \text{C-A nearest neighbors,} \end{cases}$$

taking into account only the scattering between electrons in the single-particle states $|\mathbf{k}, \uparrow\rangle$ and $|\mathbf{k}, \downarrow\rangle$

$$\hat{H}_{\text{int}} = \sum_{\mathbf{k}, \mathbf{k}', \alpha, \beta} V_{\alpha\beta}(\mathbf{k} - \mathbf{k}') \hat{c}_{\mathbf{k}, \uparrow, \alpha}^\dagger \hat{c}_{-\mathbf{k}, \downarrow, \beta}^\dagger \hat{c}_{-\mathbf{k}', \downarrow, \beta} \hat{c}_{\mathbf{k}', \uparrow, \alpha}$$



$$V_{\alpha\beta}(\mathbf{k}) = \begin{cases} U, & \alpha = \beta, \\ -V_{\alpha\beta} f(\mathbf{k}), & \alpha\beta = AB, BC, CA, \\ -V_{\alpha\beta} f^*(\mathbf{k}), & \alpha\beta = BA, CB, AC. \end{cases}$$

$$f(\mathbf{k}) = -t \sum_{\ell=1}^3 e^{i\mathbf{k} \cdot \delta_\ell}$$

引力Hubbard模型の平均場解析

平均場ハミルトニアン $c^\dagger c^\dagger c c \approx \langle c^\dagger c^\dagger \rangle c c + c^\dagger c^\dagger \langle c c \rangle$

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} H_0(\mathbf{k}) - \mu \mathbb{1} & \Delta(\mathbf{k}) \\ \Delta^\dagger(\mathbf{k}) & -H_0^\top(-\mathbf{k}) + \mu \mathbb{1} \end{pmatrix}$$

$$\Delta_{\alpha\beta}(\mathbf{k}) = \sum_{\mathbf{k}'} V_{\alpha\beta}(\mathbf{k} - \mathbf{k}') \langle \hat{c}_{-\mathbf{k}', \downarrow, \beta} \hat{c}_{\mathbf{k}', \uparrow, \alpha} \rangle$$

自己無撞着に解く

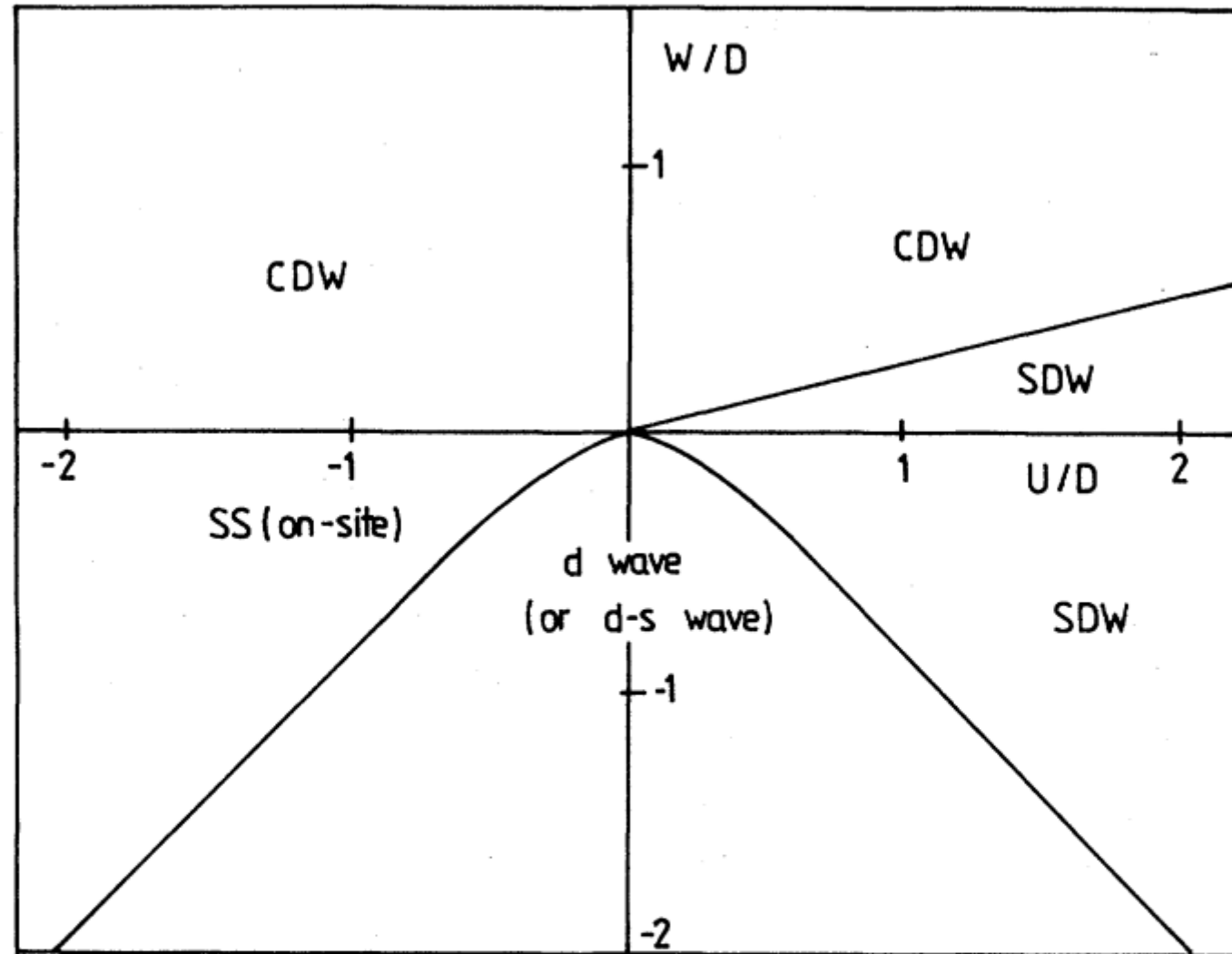
$$H_{\text{BdG}}(\mathbf{k}) \phi_{\mathbf{k}}^{(n)} = E_{\mathbf{k}}^{(n)} \phi_{\mathbf{k}}^{(n)} \quad \phi_{\mathbf{k}}^{(n)} = (\mathbf{u}_{\mathbf{k}}^{(n)}, \mathbf{v}_{\mathbf{k}}^{(n)})$$

$$\Delta_{\alpha\beta}(\mathbf{k}) = - \sum_{\mathbf{k}'} V_{\alpha\beta}(\mathbf{k} - \mathbf{k}') \sum_{n, E_n > 0} u_{\mathbf{k}', \alpha}^{(n)} v_{\mathbf{k}', \beta}^{(n)*} \tanh\left(\frac{E_{\mathbf{k}'}^{(n)}}{2T}\right)$$

引力Hubbard模型の先行研究：正方格子

Half-fillingの相図

R. Micnas *et al.*, Rev. Mod. Phys. **62**, 113 (1990).



U : オンサイト斥力(> 0)・引力(< 0)
 W : 最隣接斥力(> 0)・引力(< 0)

In Fig. 15 we give a preliminary ground-state phase diagram for $n = 1$, based on a comparison of critical temperatures for pure phases. For $U > 0$ and $W > 0$ there are only two phases possible: the CDW and the SDW, which are separated by the line $U = 4W$. For $U < 0$, the CDW and the singlet-superconducting (SS) state meet at $W = 0$. For $W < 0$ and $U < 0$ one has the SS (on-site) pairing and the d -wave pairing, whereas the SDW and the d -wave pairing are possible for $W < 0$ and $U > 0$. We should like

FIG. 15. A preliminary phase diagram of the extended Hubbard model in two dimensions and a square lattice and for the half-filled band: CDW, charge-density waves; SDW, spin-density waves; SS, singlet pairing (on-site); d -s wave, a possible mixed superconducting state.

引力Hubbard模型の先行研究：正方格子

Fillingを変えた場合(W 引力)の相図

R. Micnas *et al.*, Rev. Mod. Phys. **62**, 113 (1990).

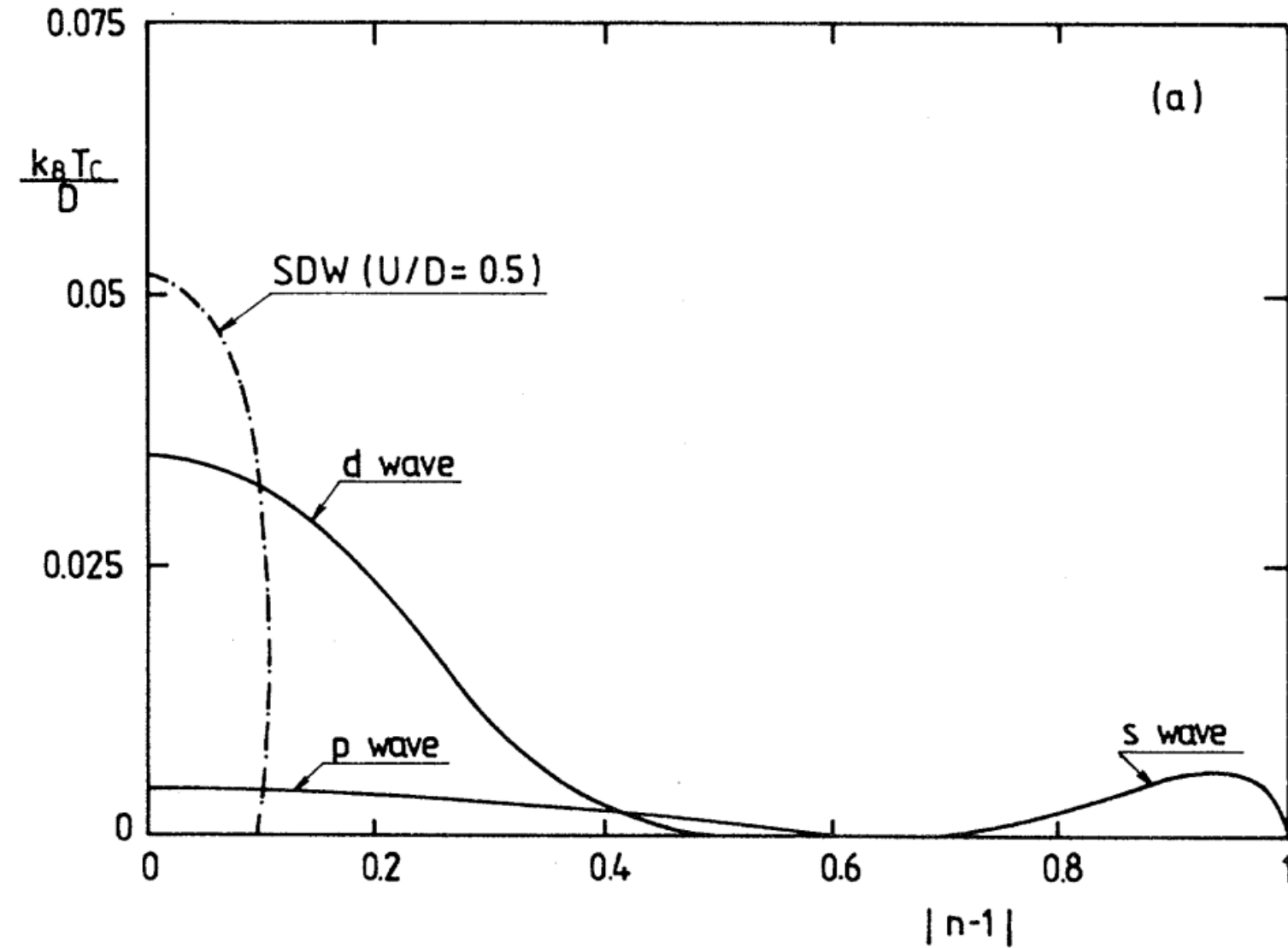


FIG. 16. T_c vs n for s -, p -, and d -wave pairings on a square lattice and with nearest-neighbor hopping. T_c for s -wave pairing is plotted for different values of U/D . Notice that T_c for p - and d -wave pairings is independent of U . (a) $|W|/D=0.25$ (T_c^s is for $U=0$), (b) $|W|/D=0.5$. The dot-dashed curves mark T_N for the spin-density wave state.

For the non-half-filled band the nature of the pairing state strongly depends on band filling. The numerical solutions for T_c versus electron density are given in Figs. 16(a) and 16(b). These plots are symmetric with respect to $n \rightarrow 2-n$, due to electron-hole symmetry. While d -wave pairing yields the highest T_c close to $n=1$, the competing p -wave and extended s -wave pairings can be stabilized for lower electron densities. T_c for s -wave pairing shows nonmonotonic behavior with rapid increase for small n , going through a maximum upon increasing n , and then vanishing above some value of n .

引力Hubbard模型の平均場解析($\alpha = 1$, dice lattice)

(a) $\alpha = 1$, $\mu/t = -1$ (1/4-filling), $T/t = 0.01$, $U/t = 4$

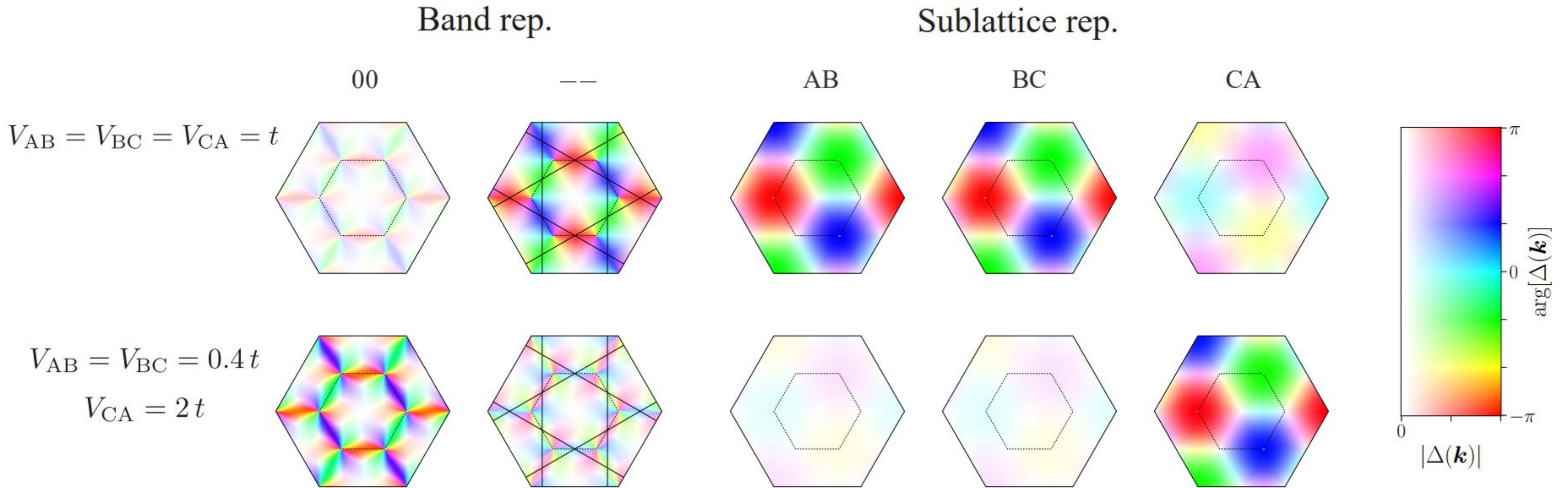
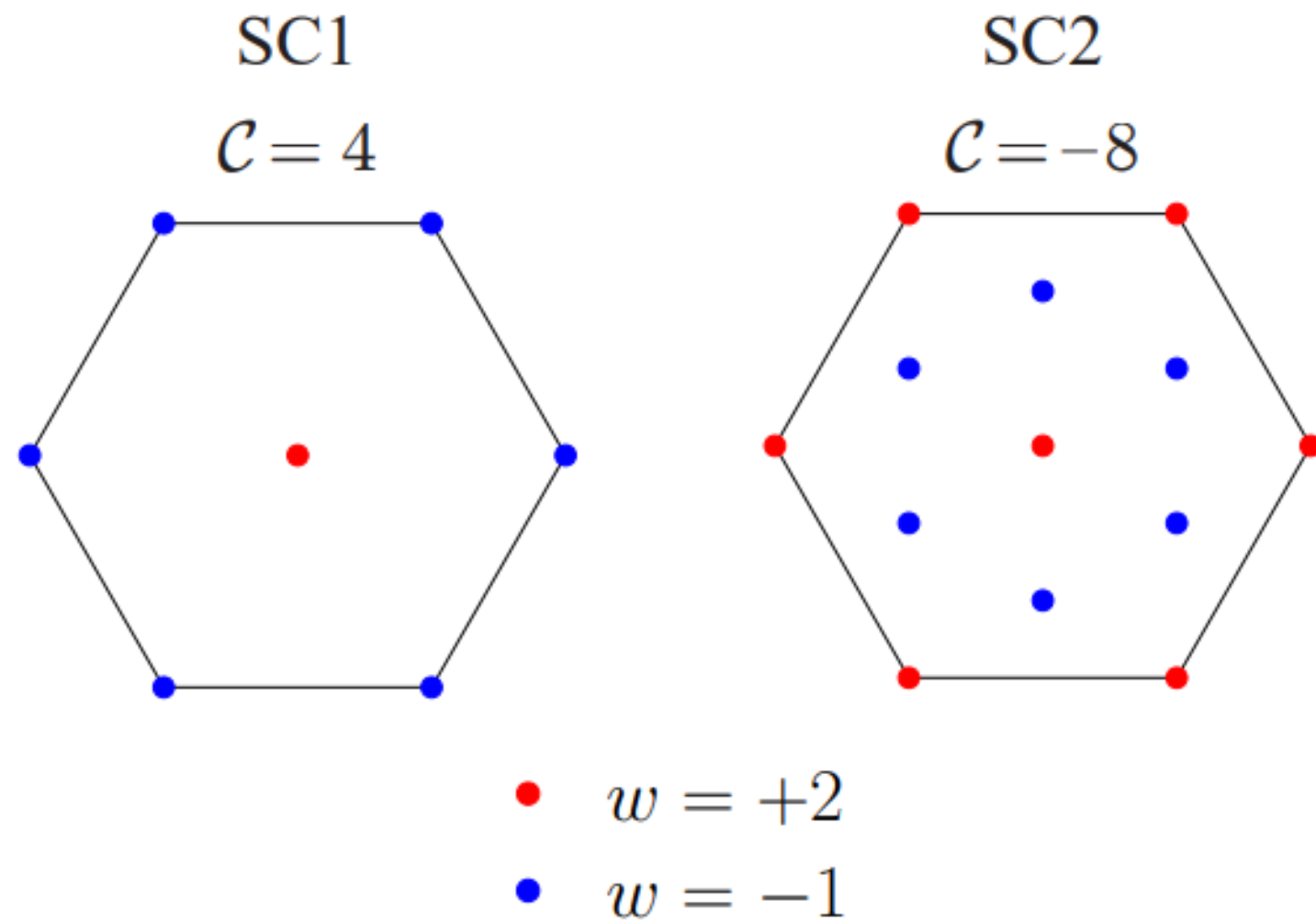


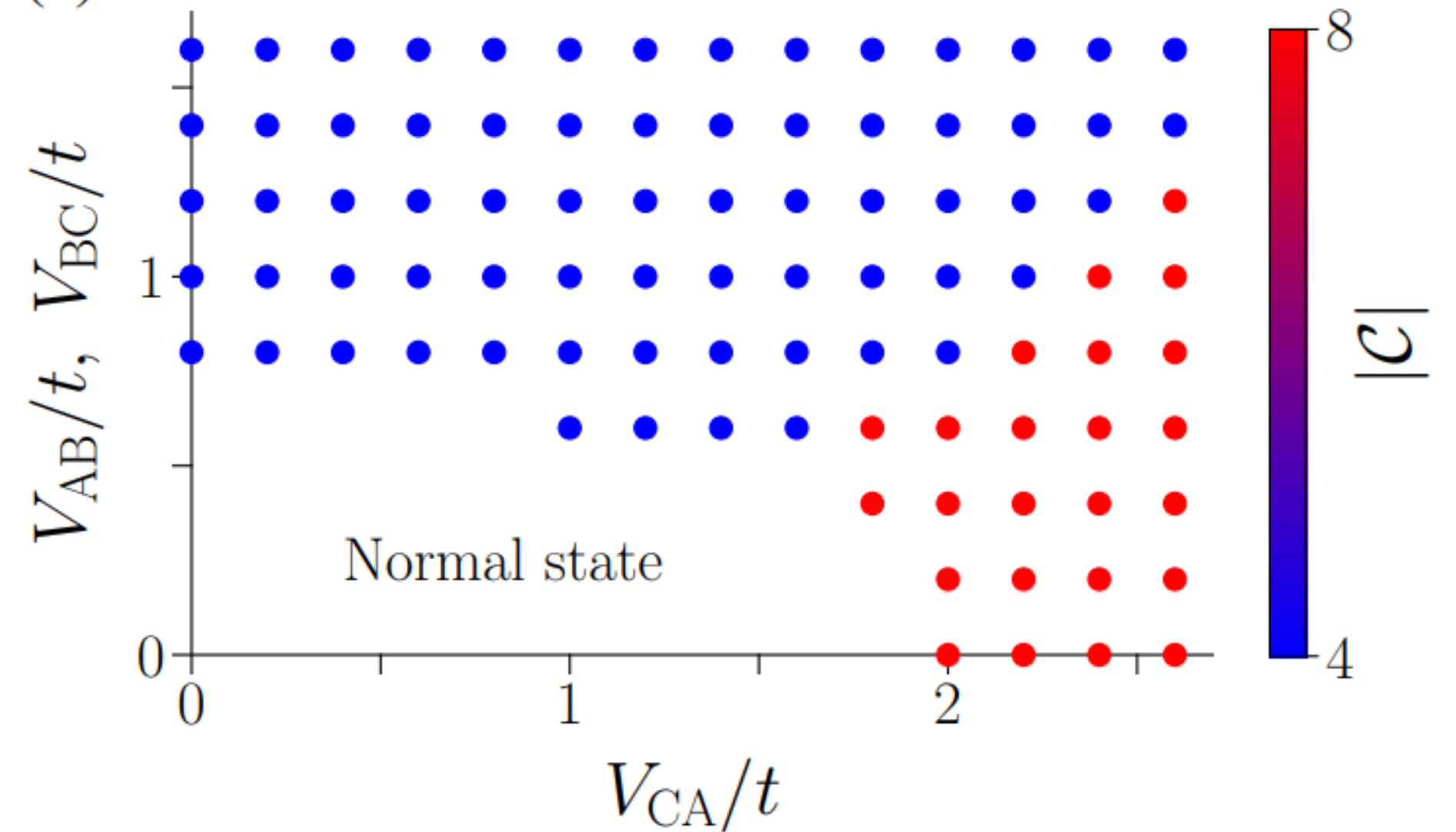
FIG. 2. (a) Band and sublattice representations of the pairing potentials $\Delta(\mathbf{k})$ obtained by self-consistently solving the BdG equation using an extended Hubbard model with nearest-neighbor attractive interactions. Two representative parameter sets of the attractions ($V_{AB} = V_{BC} = V_{CA}$ and $V_{CA} \gg V_{AB} = V_{BC}$) are shown. The region enclosed by the dashed lines denotes the first Brillouin zone (BZ), and the black solid contour in the (--) panels indicates the Fermi surface of the single-particle band.

引力Hubbard模型の平均場解析（トポロジカル相転移）

(b) 点ノードの分布と対応する巻きつき数



(c) 占有BdGバンドの総チャンネル数の相図



Total Chern number (Fukui-Hatsugai-Suzukiの方法[1]で計算)

$$\mathcal{C} = \frac{1}{2\pi i} \sum_{n=1}^3 \int_{\text{BZ}} d^2 \mathbf{k} \left(\frac{\partial A_y^{(n)}}{\partial k_x} - \frac{\partial A_x^{(n)}}{\partial k_y} \right)$$

[1] T. Fukui *et al.*, J. Phys. Soc. Jpn. 74, 1674 (2005)

引力Hubbard模型の平均場解析($\alpha = 0.5$)

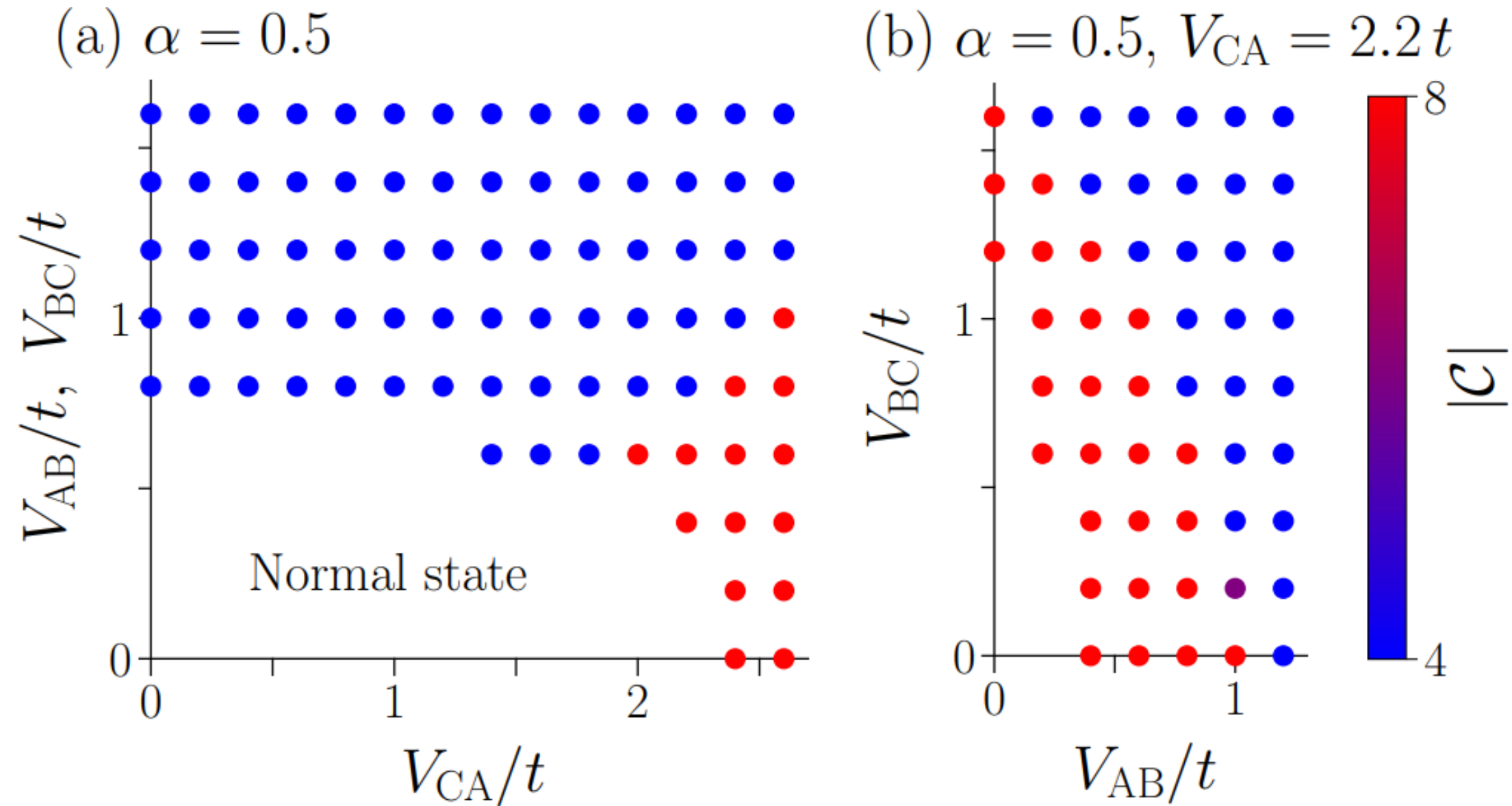
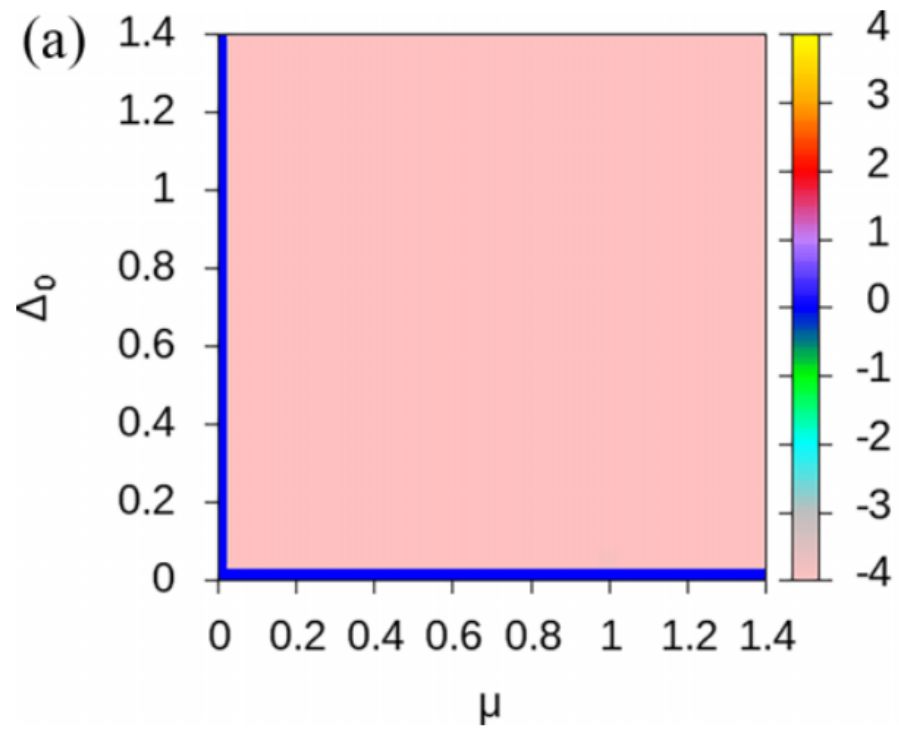


FIG. 3. Superconducting phase diagrams for (a) V_{CA} versus $V_{AB} = V_{BC}$ and (b) V_{AB} versus V_{BC} with V_{CA} fixed, calculated at $\alpha = 0.5$. All other parameters are the same as in Fig. 2.

グラフェンの場合のチャーン数

単層

$d + id'$



$p + ip'$

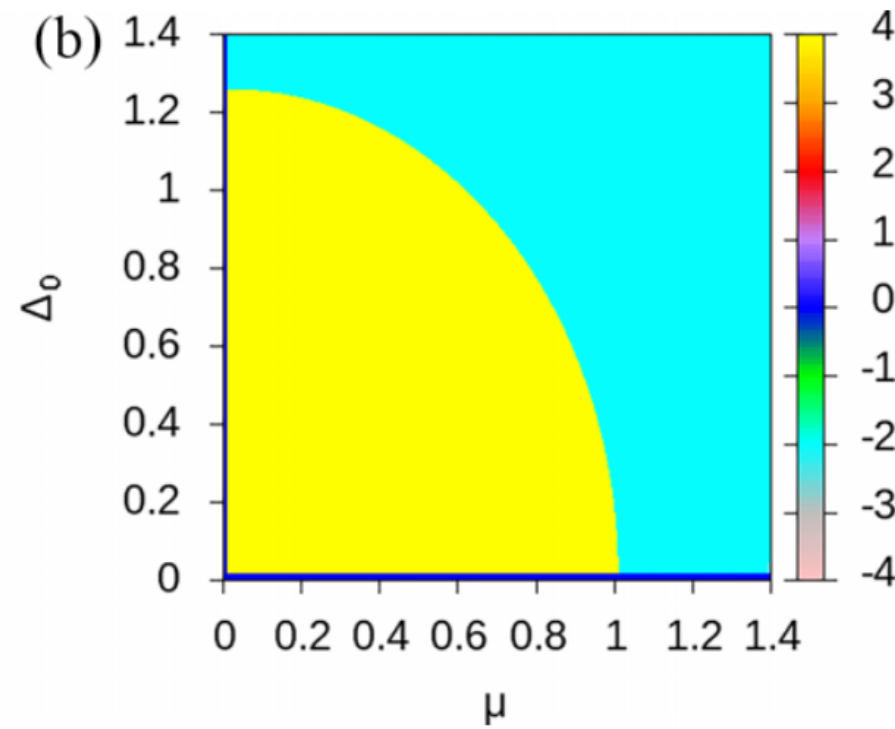
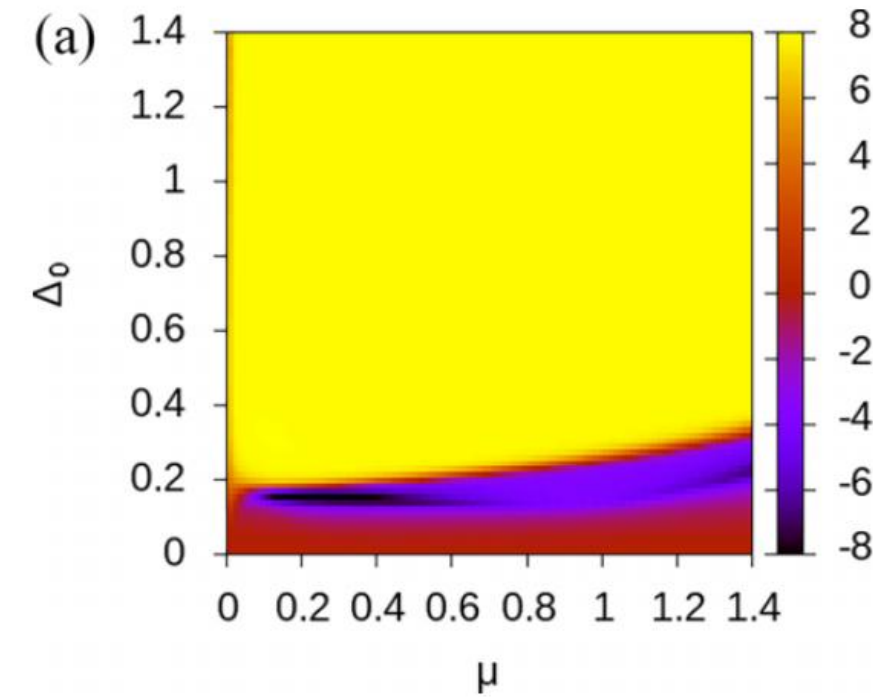


FIG. 3. Chern number $\mathcal{C}$ as a function of μ and Δ_0 for monolayer graphene for the (a) $d + id'$ - and (b) $p + ip'$ -wave states.

A. Crépieux *et al.*, Phys. Rev. B **108**, 134515 (2023)

二層

$d + id'$



$p + ip'$

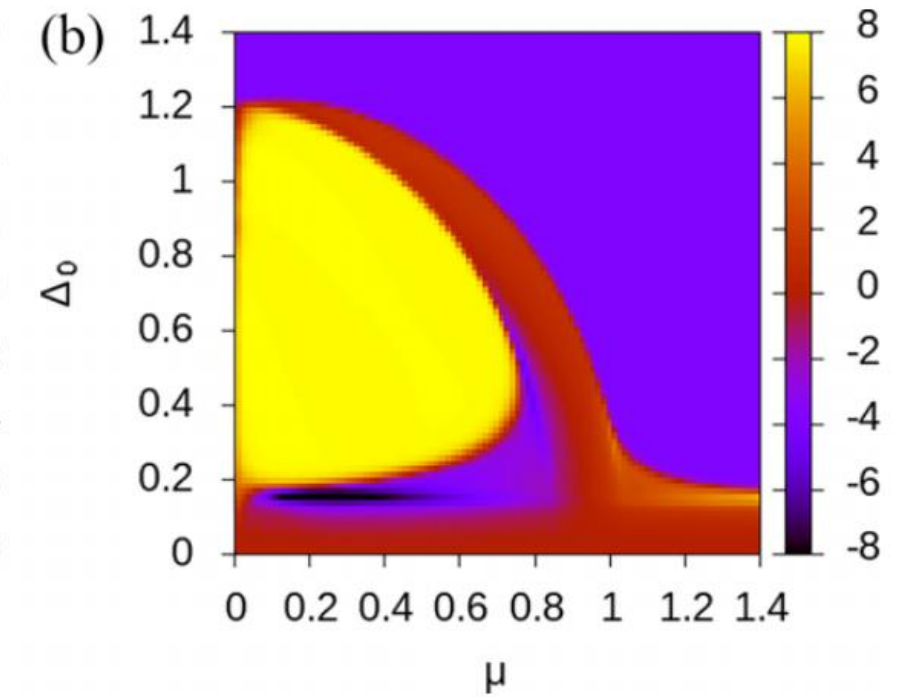


FIG. 9. Chern number $\mathcal{C}$ for AB-bilayer graphene with (a) $d + id'$ - and (b) $p + ip'$ -wave SC order parameter as a function of μ and Δ_0 at $\gamma_1 = 0.2$ and $\gamma_3 = 0$, with a phase difference $\phi = \pi$ between the SC orders in the two graphene layers.

interlayer hopping γ_1

trigonal warping γ_3

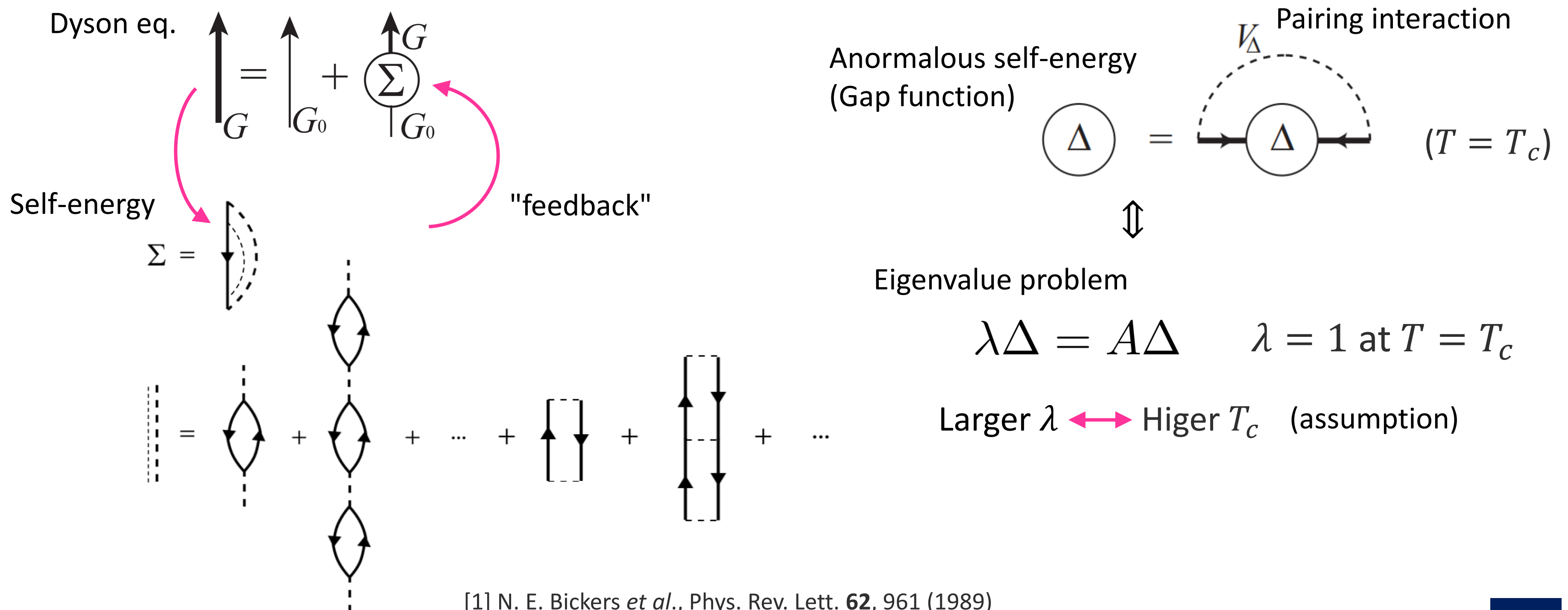
斥力Hubbard模型のスピンの揺らぎ媒介超伝導機構の検討

平均場近似では、隣接サイト間の引力を仮定した
→ 斥力相互作用のみで超伝導は可能か？

$$\hat{H}_{\text{int}} = U \sum_i \hat{n}_{i,\uparrow} \hat{n}_{i,\downarrow}$$

Hubbard模型の揺らぎ交換(FLEX)近似+線形化Eliashberg方程式

Fluctuation exchange (FLEX) approximation [1] Linearized Eliashberg equation



線形化Eliashberg方程式の固有値 λ

Condition

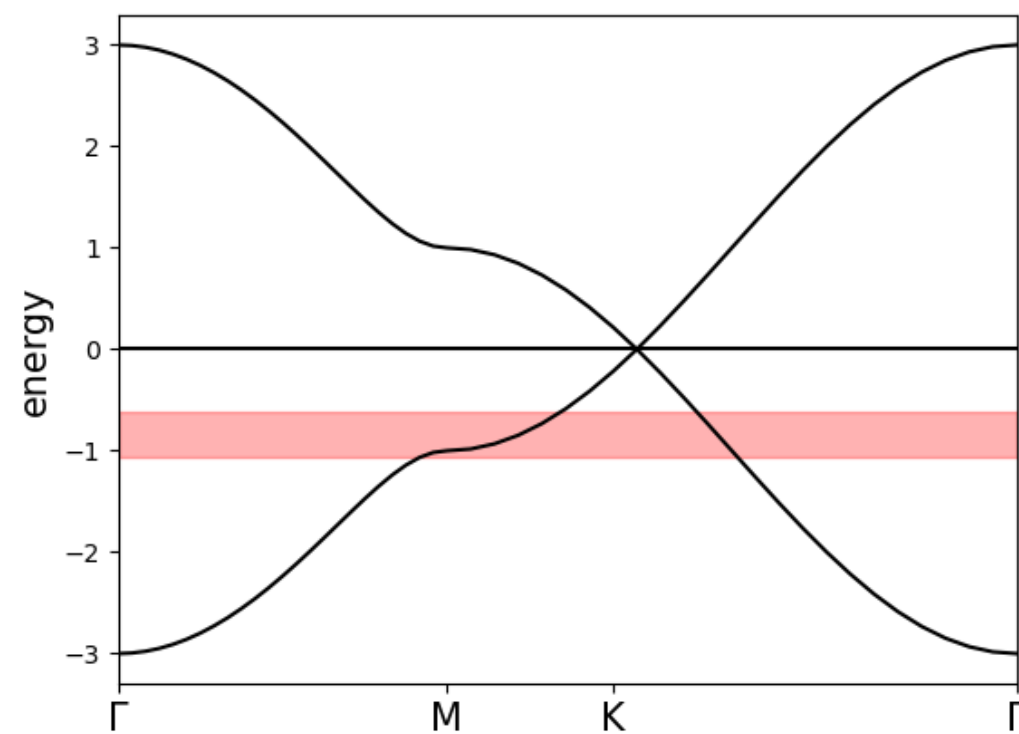
$$U = 4t$$

$$T = 0.002t$$

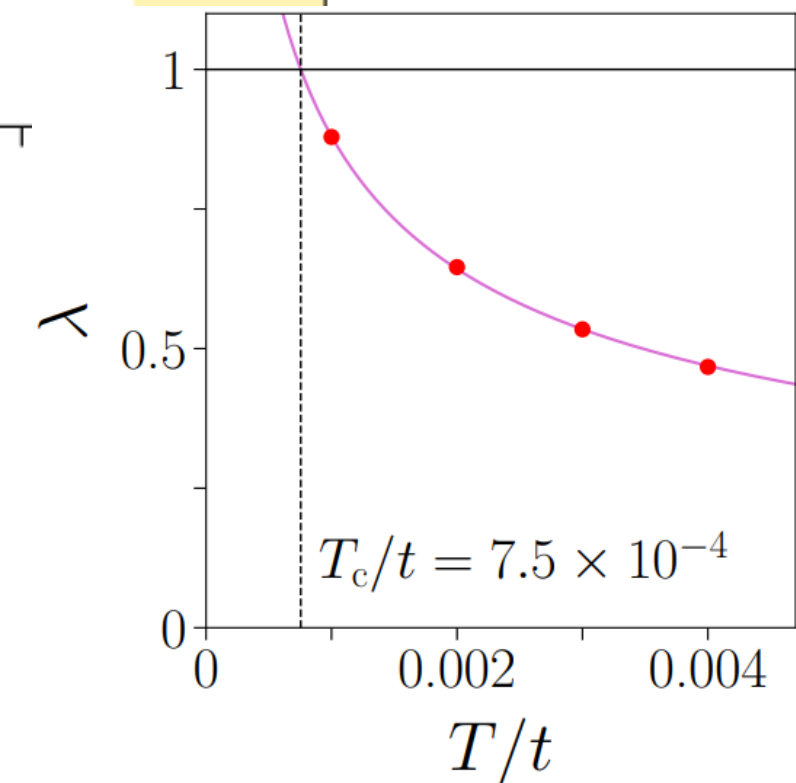
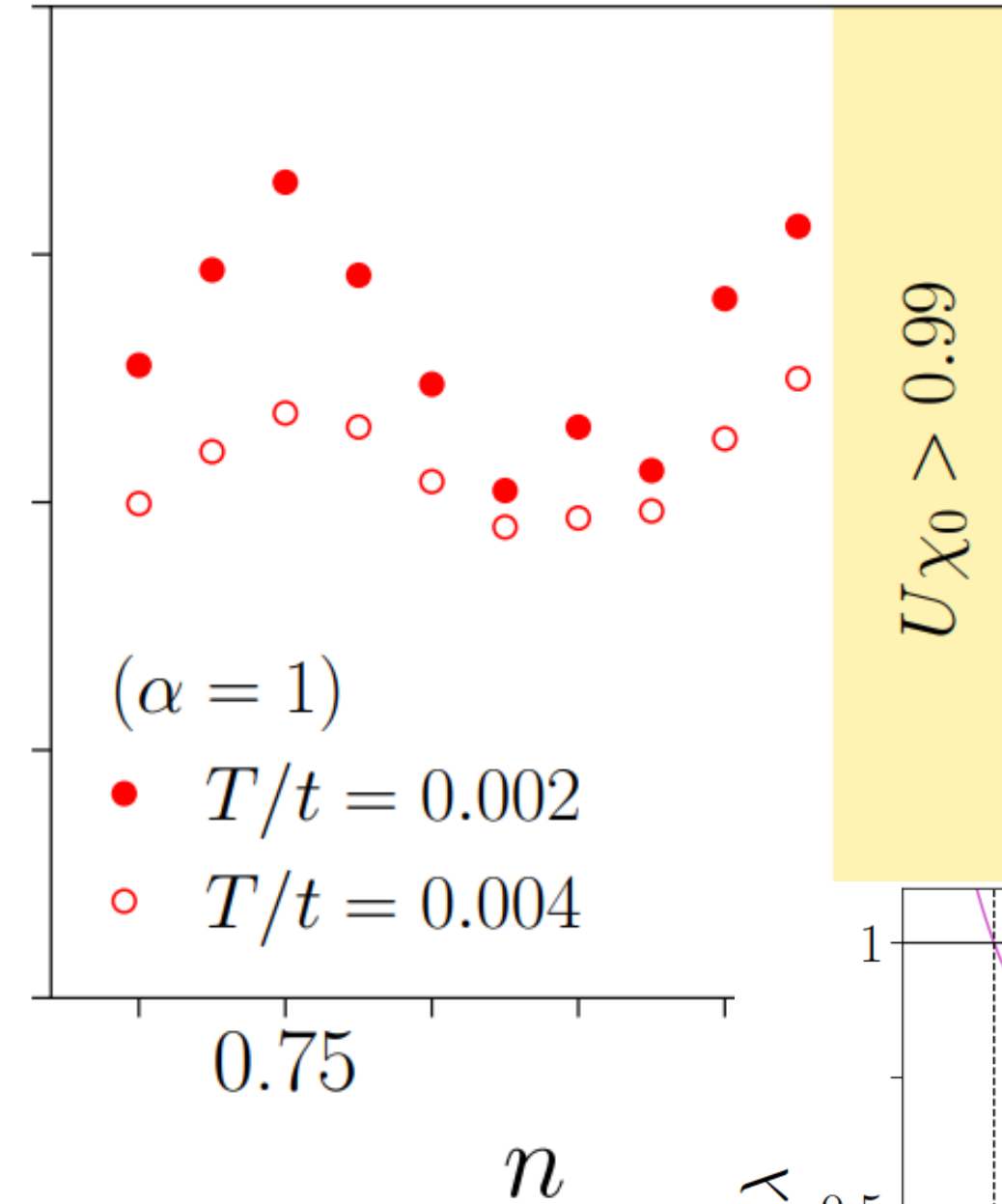
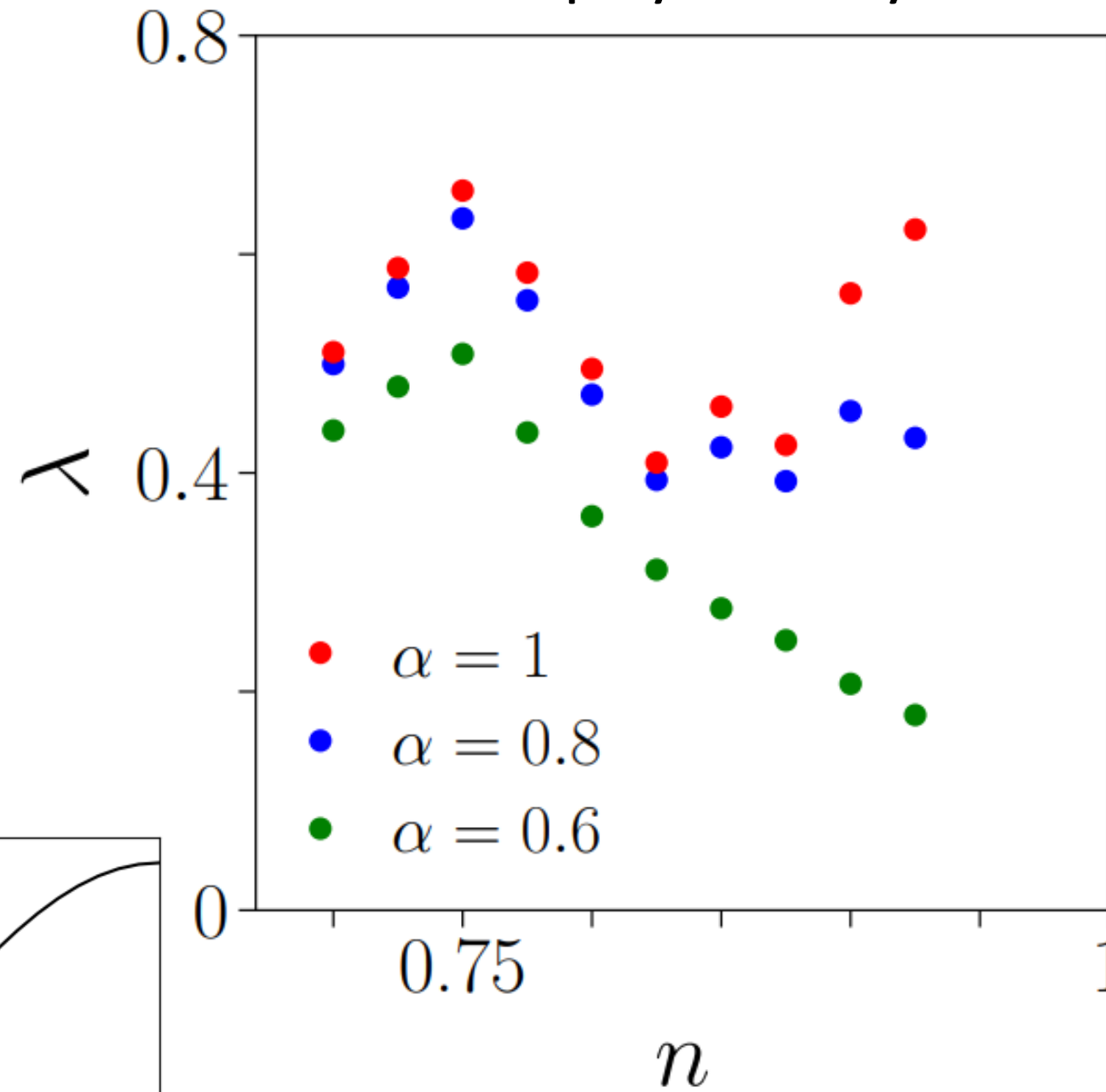
$$(N_x, N_y) = (48, 48),$$

$$N_\omega = 4096 \times 2$$

Non-interacting

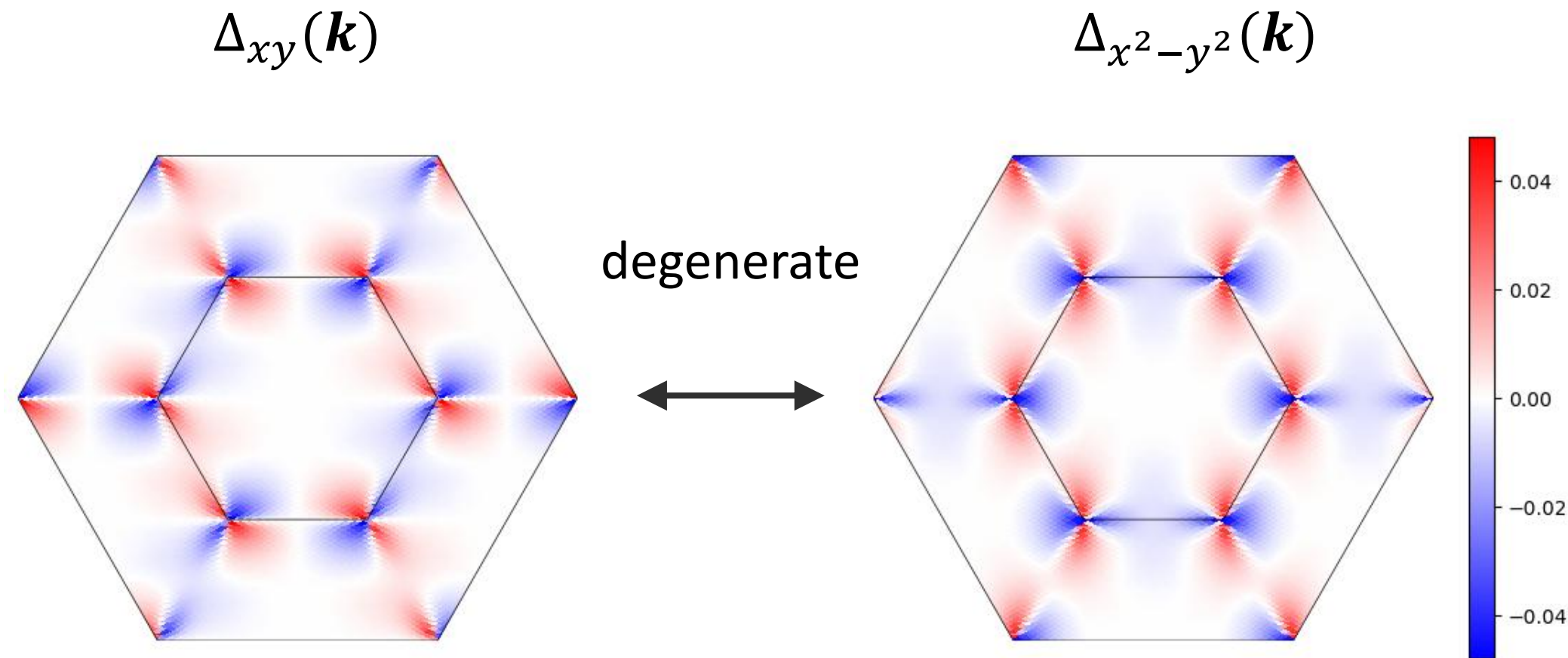


Gap symmetry: ***d*-wave** (discuss later)



FLEX+線形化Eliashberg の異常自己エネルギー（ギャップ関数）

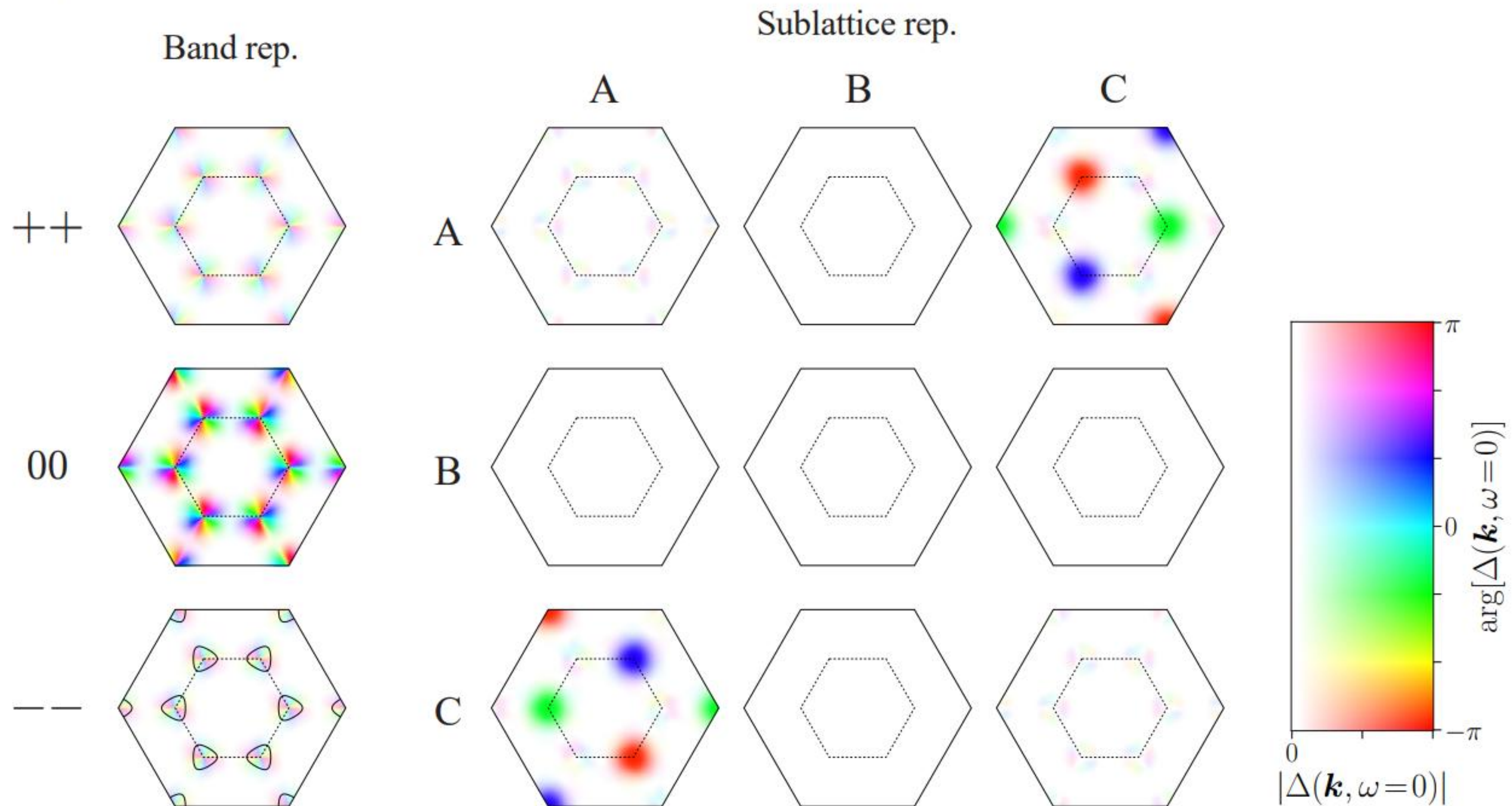
$$\alpha = 1, n = 0.9$$



mation to obtain their $\omega = 0$ components. In a hexagonal lattice structure (point group D_{6h}), the linearized Eliashberg equation gives the same T_c for any basis function belonging to a E_{2g} irreducible representation [32]. Therefore, the gap function obtained from the linearized Eliashberg equation is expressed as a linear combination of the $d_{x^2-y^2}$ and d_{xy} components. Below T_c , higher-order terms lift this degeneracy and stabilize a specific superconducting state. In general, the chiral $d + id'$ -

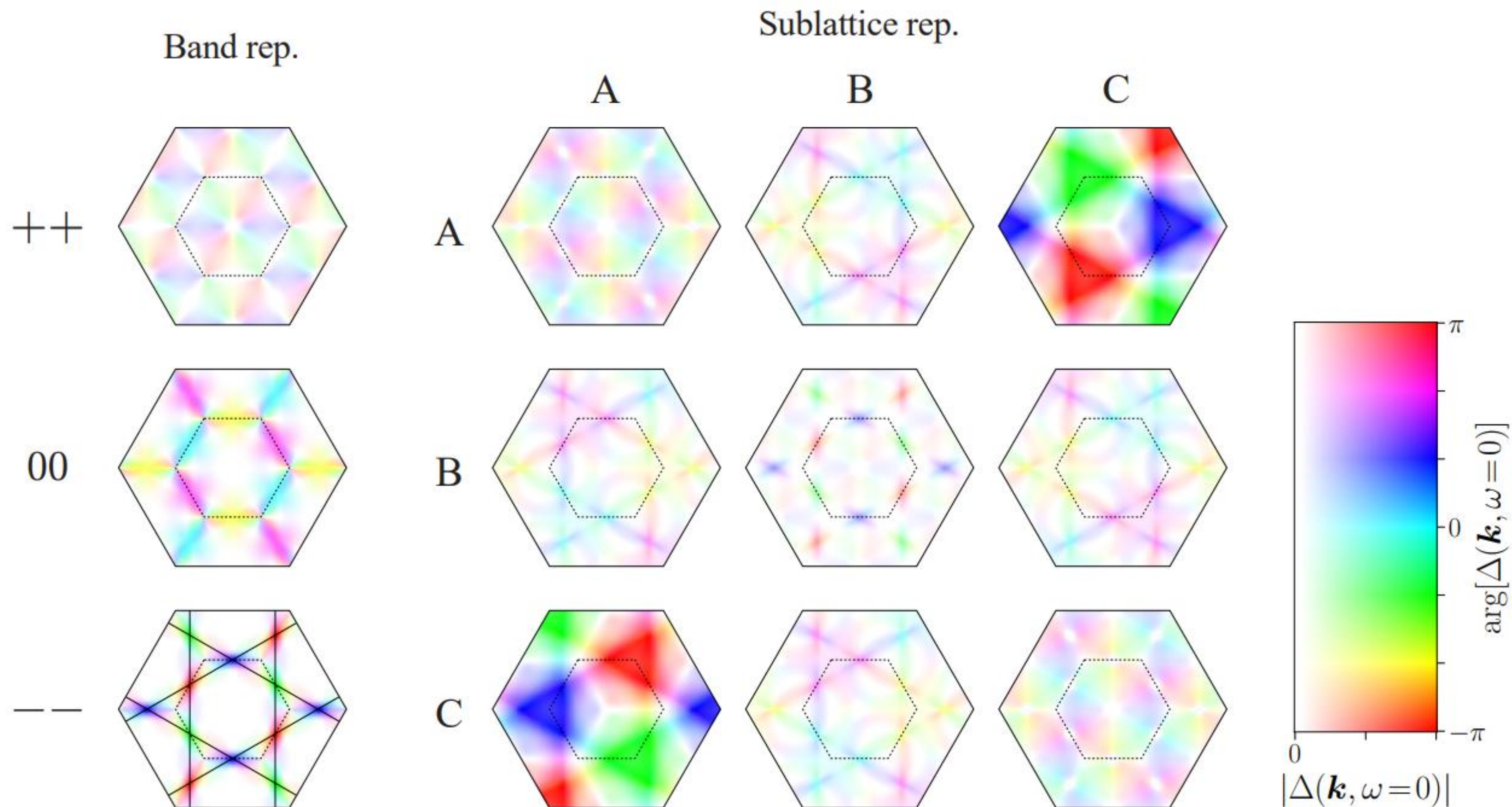
FLEX+線形化Eliashberg の異常自己エネルギー（ギャップ関数）

(b) $\alpha = 1, n = 0.9$



FLEX+線形化Eliashberg の異常自己エネルギー（ギャップ関数）

(a) $\alpha = 1, n = 0.75$



A-C間の引力の起源

スペクトル関数

動的スピン感受率

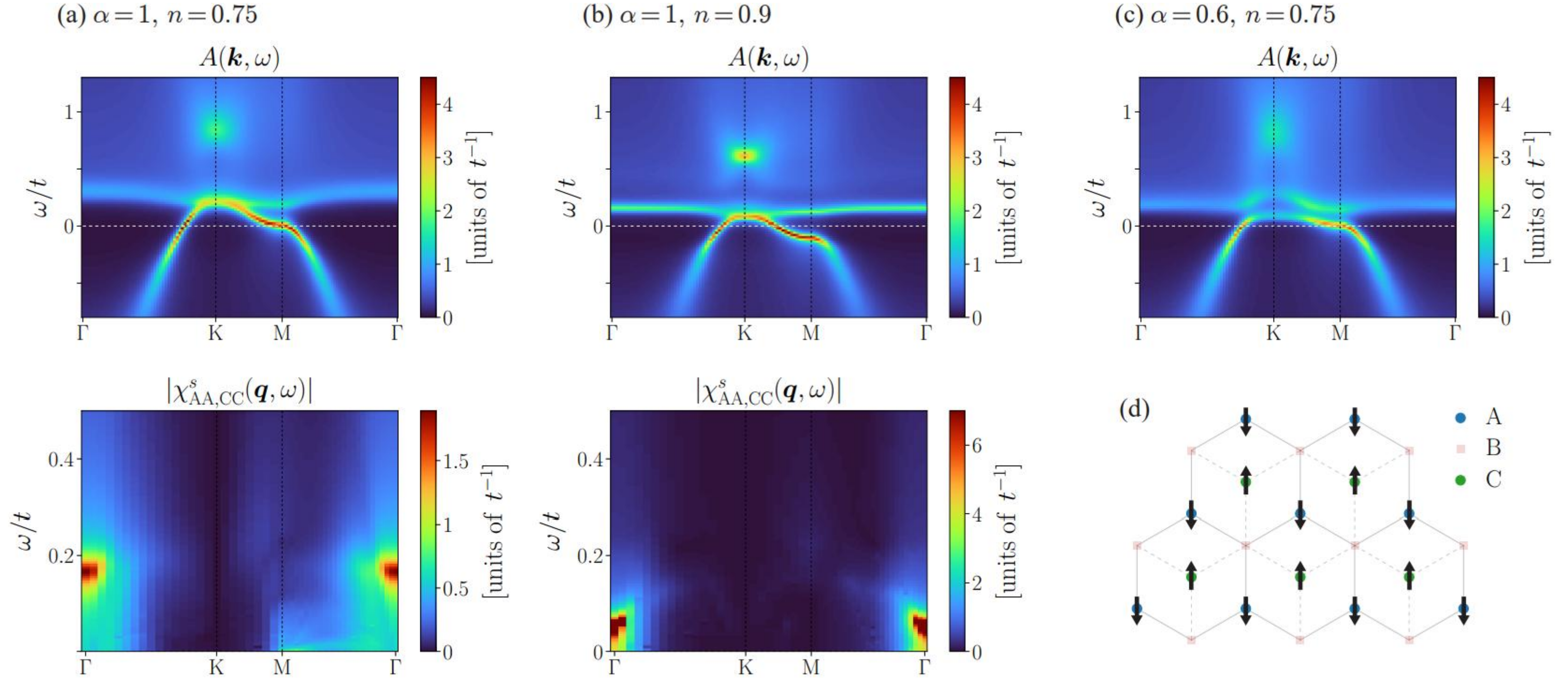
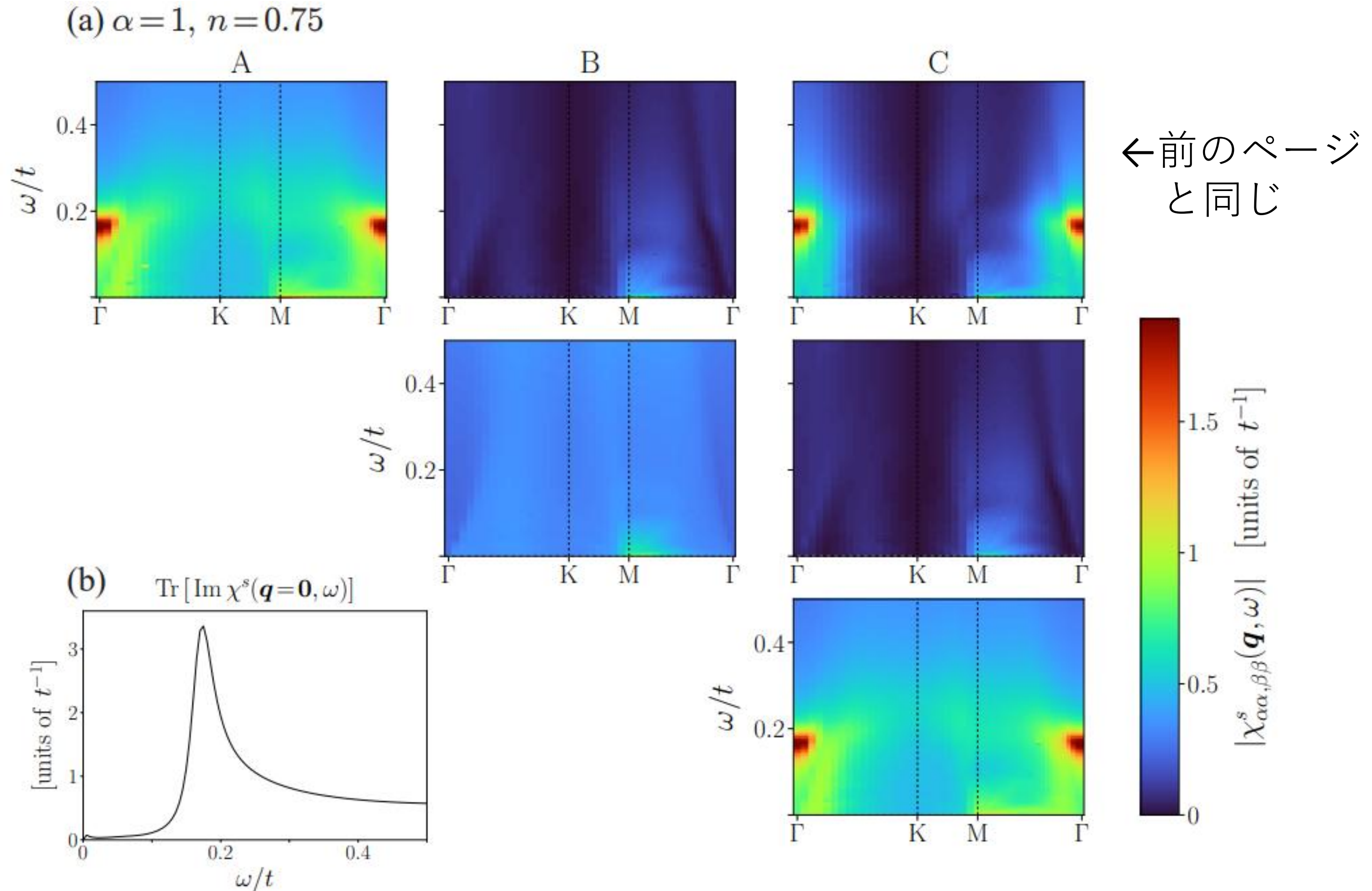


FIG. 6. Spectral functions $A(\mathbf{k}, \omega)$ (top panels) and the AC components of the dynamical spin susceptibilities $\chi_{AA,CC}^s(\mathbf{q}, \omega)$ (bottom panels) for $\alpha = 1$ and band fillings (a) $n = 0.75$ and (b) $n = 0.9$. The spectral function for $\alpha = 0.6$ and $n = 0.75$ is shown in (c) for comparison. In the dynamical spin susceptibility, the largest spectral weight appears at finite energy near $\mathbf{q} = (0, 0)$. (d) Schematic illustration of the real-space spin configuration corresponding to the $\mathbf{q} = \mathbf{0}$ spin fluctuation between sites A and C. Note that each arrow does not represent a single electron.

スピン感受率の全行列要素

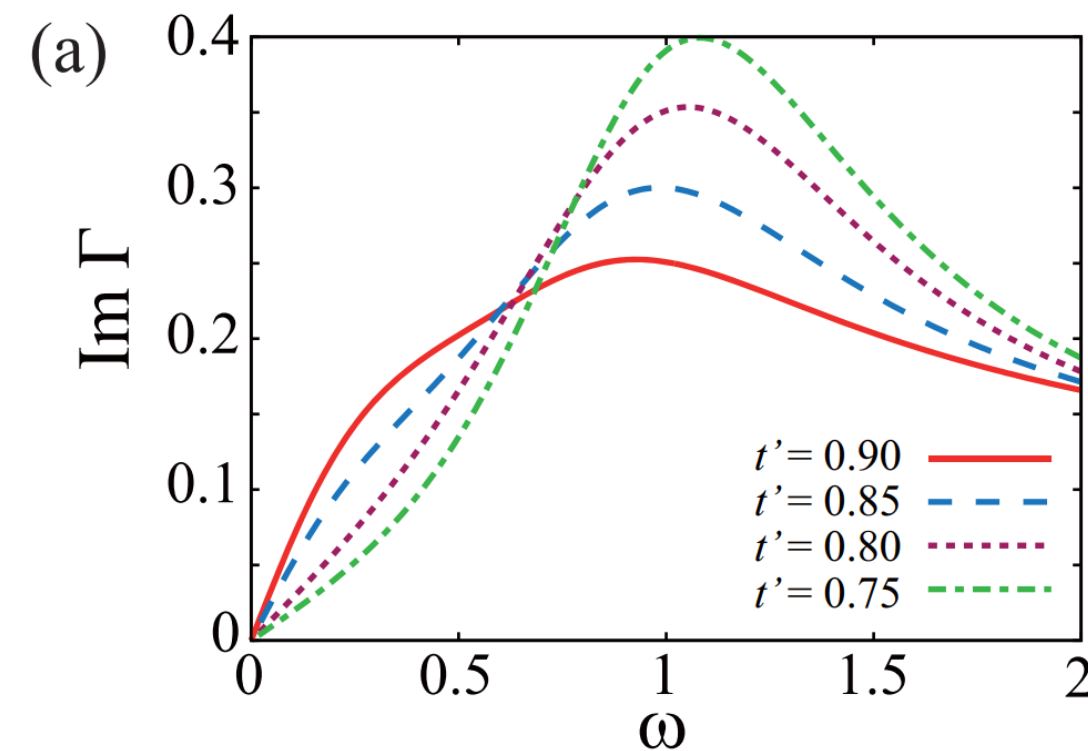


鉄系超伝導に対する議論

- [46] F. Wang, F. Yang, M. Gao, Z.-Y. Lu, T. Xiang, and D.-H. Lee, The electron pairing of $K_x\text{Fe}_{2-y}\text{Se}_2$, [Europhys. Lett. **93**, 57003 \(2011\)](#).
- [47] Y. Bang, A shadow gap in the over-doped $(\text{Ba}_{1-x}\text{K}_x)\text{Fe}_2\text{As}_2$ compound, [New J. Physics **16**, 023029 \(2014\)](#).
- [48] Y. Bang, Pairing mechanism of heavily electron doped FeSe systems: dynamical tuning of the pairing cutoff energy, [New J. Phys. **18**, 113054 \(2016\)](#).
- [49] X. Chen, S. Maiti, A. Linscheid, and P. J. Hirschfeld, Electron pairing in the presence of incipient bands in iron-based superconductors, [Phys. Rev. B **92**, 224514 \(2015\)](#).

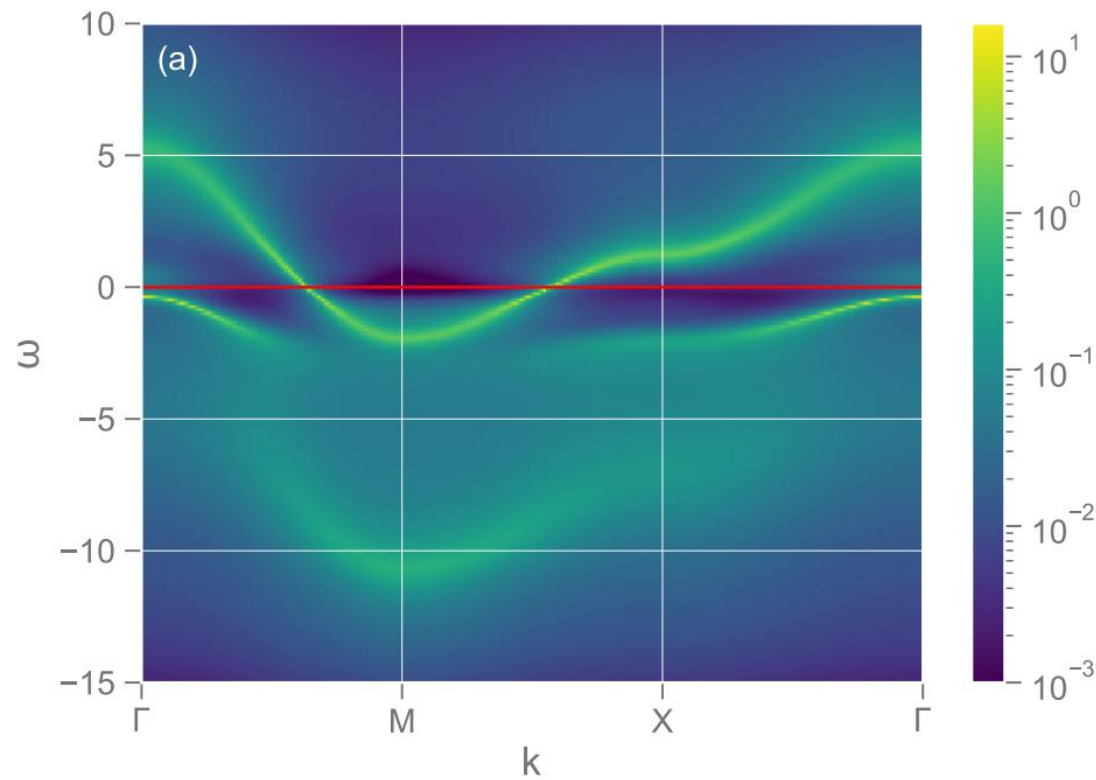
二層系などincipient bandのある系

- [55] M. Nakata, D. Ogura, H. Usui, and K. Kuroki, Finite-energy spin fluctuations as a pairing glue in systems with coexisting electron and hole bands, [Phys. Rev. B **95**, 214509 \(2017\)](#).
- [56] D. Ogura, H. Aoki, and K. Kuroki, Possible high- T_c superconductivity due to incipient narrow bands originating from hidden ladders in Ruddlesden-Popper compounds, [Phys. Rev. B **96**, 184513 \(2017\)](#).
- [59] T. A. Maier, V. Mishra, G. Balduzzi, and D. J. Scalapino, Effective pairing interaction in a system with an incipient band, [Phys. Rev. B **99**, 140504 \(2019\)](#).



有限エネルギーのスピン揺らぎ

- [59] T. A. Maier, V. Mishra, G. Balduzzi, and D. J. Scalapino, Effective pairing interaction in a system with an incipient band, *Phys. Rev. B* **99**, 140504 (2019).



(a) Single-particle spectral weight $A(\mathbf{k}, \omega)$ at $T = 0.2$ for $t_{\perp} = 2.5$, $n = 1.15$, and $U = 8$

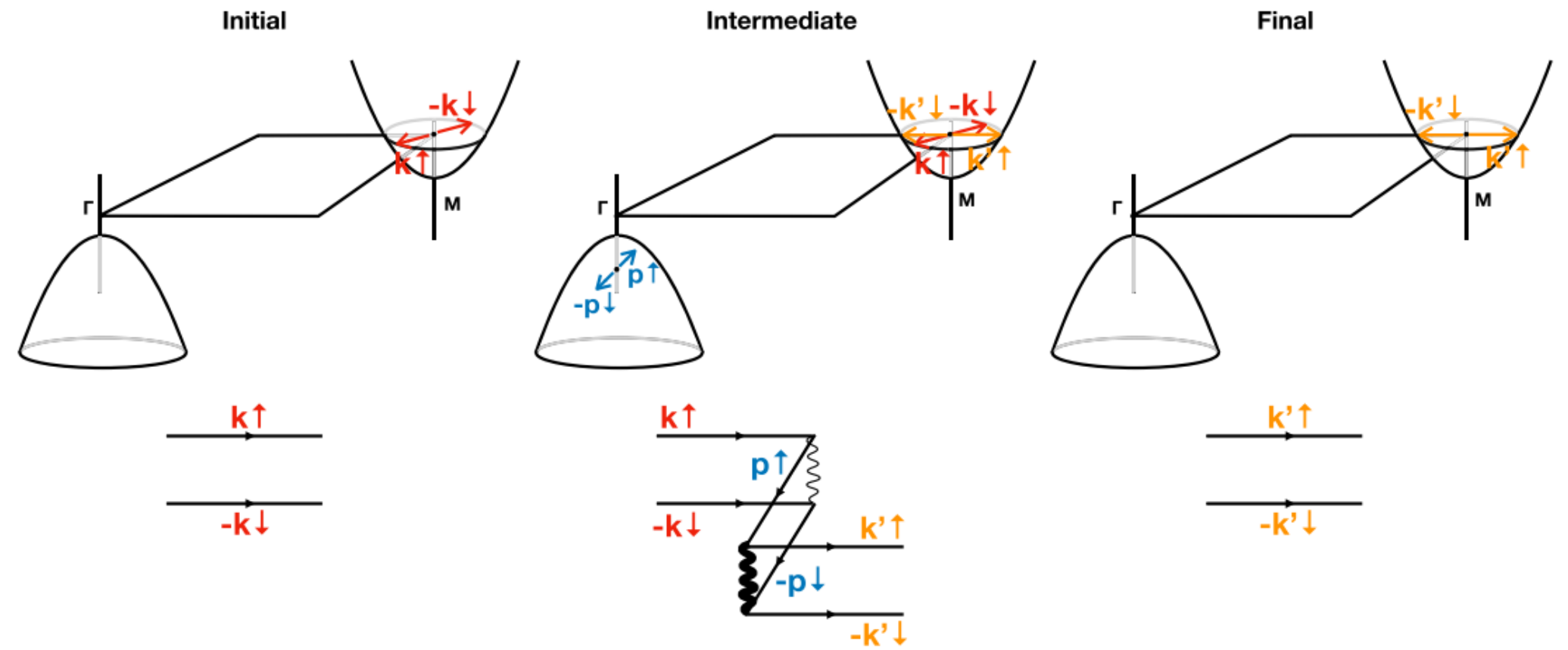


FIG. 3. Schematic illustration of the effective intermediate state contribution to the Γ_2 pairing vertex. The Feynman diagram for Γ_2 shown in the lower part of the figure involves the intermediate states illustrated above it. The thin interaction line denotes the $\Gamma_{0\pi}^0$ fully irreducible vertex and the thicker interaction line the $\Gamma_{\pi 0}$ vertex, which is only irreducible in the $k_z = 0$ two-particle channel.